
Rethinking Convergence in MoE Training: The Role of Routing Sparsity

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Abstract

In Mixture-of-Experts (MoE) training, sparse routing, i.e., activating only the top- K experts per token, is essential for balancing convergence speed and computational cost. However, existing works typically choose K empirically, without theoretical guidance. To address this gap, we characterize the convergence behavior of MoE training using stochastic optimization theory. Specifically, we derive a convergence upper bound of $\mathcal{O}\left(\frac{1+M/K}{\sqrt{T}}\right)$, where T is the number of training iterations and M is the total number of experts per MoE layer. This result guarantees convergence and shows that increasing K can accelerate training. By further fixing the total computational budget R (in FLOPs), we obtain a refined bound of $\mathcal{O}\left(\sqrt{\frac{K}{R}} + \frac{M}{\sqrt{KR}}\right)$, which is convex with respect to K and implies the existence of an optimal $K^* \in [1, M]$ that achieves the best convergence performance. Extensive experiments validate our theoretical analysis under diverse settings.

1. Introduction

Recently, Mixture-of-Experts (MoE) architectures (Jacobs et al., 1991; Jordan & Jacobs, 1994) have attracted significant attention in large-scale deep learning due to their superior computational efficiency. Benefiting from a favorable efficiency-performance trade-off, MoE models have been successfully applied across a broad range of domains, including natural language processing (Vaswani et al., 2017;

Brown et al., 2020; Du et al., 2022; Ruiz et al., 2021b), computer vision (Ruiz et al., 2021a; Liu et al., 2021b), and multimodal learning (Zhang et al., 2021; Zhou et al., 2022a; Lin et al., 2026).

Empirical studies have shown that MoE models can achieve performance comparable to dense models with the same parameter scale (Lepikhin et al., 2021; Fedus et al., 2022; Wei et al., 2024), while significantly reducing computational cost through sparse routing. Specifically, MoE training activates only a subset of experts (Shazeer et al., 2017; He et al., 2021; Zhou et al., 2022b) for each input token. An MoE model typically consists of a gating network (i.e., gate) and multiple experts, each specializing in different functionalities. For each token, the gate assigns scores to all experts and routes the token to the top- K experts with the highest scores (top- K routing, where K is a critical hyperparameter). The outputs of the selected experts are then weighted by the corresponding gating scores and aggregated to produce the final MoE output.

Existing works study the role of K primarily through empirical observations; for example, (Ramachandran & Le, 2019; Qin et al., 2025) show that increasing K can accelerate convergence. However, a theoretical foundation for selecting K remains lacking. Existing work (Fedus et al., 2022; Clark et al., 2022; Lepikhin et al., 2021; Rajbhandari et al., 2022) typically chooses K heuristically or by general rule of thumb, which can lead to suboptimal performance. If K is too small (e.g., $K = 1$), computational cost per token is minimized, but convergence may slow dramatically or even diverge. Conversely, if K is too large (e.g., $K = M$), convergence is ensured at the expense of prohibitively high computational cost. Moreover, fixing K without theoretical guidance lacks the flexibility to adapt to different models and computational budgets.

These limitations highlight the need for a theoretical analysis of the relationship between routing sparsity, convergence, and efficiency in MoE training. To this end, this paper investigates the impact of sparse routing on MoE training from a theoretical perspective. The main contributions of this paper are summarized as follows.

- We provide one of the first theoretical analysis of the convergence upper bound of MoE training, revealing

the relationship between routing sparsity, convergence behavior, and computational cost via stochastic optimization theory. Furthermore, we extend the analysis to characterize the convergence under given fixed total computational budget.

- Given sufficient computational budget, we show that the convergence upper bound scales as $\mathcal{O}\left(\frac{1+M/K}{\sqrt{T}}\right)$, where T is the number of training iterations and M is the number of experts per MoE layer. In this context, the result guarantees convergence and indicates that activating more experts per token can accelerate convergence.
- Under a fixed total computational budget R (in FLOPs), we prove that the convergence upper bound scales as $\mathcal{O}\left(\sqrt{\frac{K}{R}} + \frac{M}{\sqrt{KR}}\right)$, which is convex with respect to K . This result implies the existence of an optimal $K^* \in [1, M]$ that achieves the best convergence performance given any R .
- Extensive experiments on image classification and language generation corroborate our theory. Results show that, with sufficient computational budget, convergence speed increases with K but exhibits diminishing returns; under limited computational budgets, convergence performance first improves and then degrades as K increases.

2. Related Works

Existing theoretical studies on MoE have advanced along several complementary directions. Early work (Jordan & Xu, 1995) analyzes the convergence rate of MoE under the expectation-maximization algorithm that minimizes the maximum-likelihood estimation. With the rise of deep learning (DL), (Chen et al., 2022) provides the first convergence analysis of MoE in DL, and identifies that intrinsic cluster structures in the data and non-linear properties of experts are key factors underlying MoE’s effectiveness. It further shows that the gate can learn cluster-center representations, enabling MoE to solve clustered classification problems that are intractable for single-expert models. Based on (Chen et al., 2022), (Chowdhury et al., 2023) investigates the generalization performance of patch-level MoE (pMoE) on vision tasks, and demonstrates that discriminative routing can reduce sample complexity by filtering irrelevant image patches. A separate line of work focuses on the generalization and convergence properties of MoE under different routing mechanisms and expert architectures. For instance, (Nguyen et al., 2024a) investigates the convergence rate of softmax-gated multinomial logistic MoE, and proposes a modified routing scheme that mitigates harmful gate-expert interactions to improve the convergence rate. In regression settings, (Nguyen et al., 2024b) analyzes least-squares

estimators of MoE, and shows that feed-forward experts with sigmoid or tanh activations achieve faster estimation rates than polynomial-based experts. For Gaussian-gated MoE, (Nguyen et al., 2024c) investigates the convergence of maximum-likelihood estimation by using Voronoi-type loss functions, and characterizes distinct convergence behaviors under different routing mechanisms.

Beyond the aforementioned fundamental analysis, recent work has explored MoE theory in application contexts. For instance, (Li et al., 2025) provides the first theoretical characterization of MoE in continual learning (CL) in terms of generalization and convergence performance. It proves that MoE enables expert task specialization and router-based task selection compared with dense models, and identifies the necessity to terminate gating network updates for better convergence. Furthermore, it also derives explicit expressions for expected forgetting/generalization error for CL. In the context of mobile edge computing (MEC), (Li & Duan, 2025) pioneers MoE theory to mitigate CL overfitting/forgetting. It treats each MEC server as an expert and designs an adaptive routing mechanism for unknown arrival tasks. Moreover, it also analyzes the minimum number of experts required, and demonstrates the significant reduction of long-term generalization error compared with traditional CL for MEC.

3. Problem Formulation

In this section, let us first introduce the forward computation workflow of MoE in detail, and then describe the SGD algorithm for training MoE.

3.1. Forward Computation Workflow of MoE

The forward computation workflow of MoE consists of the following steps.

(1) Gating and Routing. For each token $\mathbf{x}$, the gating computes a vector of routing scores $g(\mathbf{w}_g, \mathbf{x}) \in \mathbb{R}^M$, where $\mathbf{w}_g$ denotes the parameter of the gate and M denotes the total number of experts. Let us define the index set of all experts as $\mathcal{E} = \{1, 2, \dots, m, \dots, M\}$. Correspondingly, the m -th entry of $g(\mathbf{w}_g, \mathbf{x})$ (i.e., $g(\mathbf{w}_g, \mathbf{x})[m] \in \mathbb{R}$) denotes the routing scores of the m -th expert. Based on these scores, the experts with the top- K highest routing scores are activated, and the gate produces normalized routing weights that determine how much each activated expert contributes to the final output:

$$\pi(\mathbf{w}_g, \mathbf{x}, \mathcal{A})[m] = \begin{cases} \frac{\exp(g(\mathbf{w}_g, \mathbf{x})[m])}{\sum_{m' \in \mathcal{A}} \exp(g(\mathbf{w}_g, \mathbf{x})[m'])}, & \text{if } m \in \mathcal{A}, \\ 0, & \text{if } m \notin \mathcal{A}, \end{cases} \quad (1)$$

where $\mathcal{A} \subseteq \mathcal{E}$ denotes the index set of activated experts with the token $\mathbf{x}$.

(2) Expert Computation. The activated experts independently process the routed token. Let us define the parameter vector of the experts as $\mathbf{w}_m, \forall m \in \mathcal{E}$, and for the token $\mathbf{x}$, the output of the m -th expert is defined as $E(\mathbf{w}_m, \mathbf{x}_i) \in \mathbb{R}^{d_{\text{out}}}$, where d_{out} denotes the number of dimensions of the expert output.

(3) Expert Outputs Aggregation. The outputs of the activated experts are aggregated to generate the final MoE output for the token $\mathbf{x}$:

$$\text{MoE}(\mathbf{w}_g, \mathbf{x}, \mathcal{A}) = \sum_{m \in \mathcal{A}} \pi(\mathbf{w}_g, \mathbf{x}, \mathcal{A})[m] E(\mathbf{w}_m, \mathbf{x}). \quad (2)$$

3.2. SGD for Training MoE

Due to sparse routing, the training mode of MoE is different from that of dense models. Let us consider a deep multi-layer MoE-based model consisting of N MoE layers stacked in sequence, wherein the output of each MoE layer is fed as the input to the subsequent MoE layer for further processing. Moreover, we consider homogeneous model architectures for all MoE layers and each MoE layer comprises a single gate and M experts. Let us define the layer index set as $\mathcal{N} = \{1, 2, \dots, n, \dots, N\}$, the expert index set for each layer as $\mathcal{M} = \{1, 2, \dots, m, \dots, M\}$, and the index set of all experts in the MoE-based model as $\mathcal{E} = \{(n, m)\}_{n \in \mathcal{N}, m \in \mathcal{M}}$, where the (n, m) -th expert denotes the m -th expert in the n -th layer. In addition to the gates and experts, the modern MoE-based model also includes non-MoE components (e.g., the self-attention modules of MoE-based Transformer models (Vaswani et al., 2017)).

Given the training dataset $\mathcal{D}$ with size of D , the training objective of the MoE-based model is to learn the optimal model parameters to minimize the average loss function over the training dataset:

$$\min_{\mathbf{w}} F(\mathbf{w}, \mathcal{D}) = \min_{\mathbf{w}} \frac{1}{D} \sum_{\mathbf{x}_i \in \mathcal{D}} f(\mathbf{w}, \mathbf{x}_i, \mathcal{A}_i). \quad (3)$$

where $\mathbf{x}_i$ denotes the i -th token in $\mathcal{D}$, $\mathcal{A}_i$ denotes the index set of the activated experts for the token $\mathbf{x}_i$, and $f(\cdot)$ denotes the loss function with respect to the model parameters $\mathbf{w}$, the token $\mathbf{x}_i$, and the corresponding index set of the activated experts $\mathcal{A}_i$. Moreover, $\mathbf{w}$ denotes the concatenated vector of all parameters of the MoE-based model:

$$\mathbf{w} = \text{concat}(\mathbf{w}_s, \mathbf{w}_{g,1}, \mathbf{w}_{(1,1)}, \dots, \mathbf{w}_{(1,M)}, \dots, \mathbf{w}_{g,2}, \mathbf{w}_{(2,1)}, \dots, \mathbf{w}_{(2,M)}, \dots, \mathbf{w}_{g,N}, \mathbf{w}_{(N,1)}, \dots, \mathbf{w}_{(N,M)}), \quad (4)$$

where $\text{concat}(\cdot)$ denotes the vector concatenation operator, $\mathbf{w}_s$ denotes the parameter vector of non-MoE components, $\mathbf{w}_{g,n}$ denotes the parameter vector of the gate in the n -th layer, and $\mathbf{w}_{(n,m)}$ denotes the parameter vector of the (n, m) -th expert.

To solve the optimization problem (3), a widely adopted approach is the SGD algorithm (Ketikar, 2017), which iteratively updates model parameters by computing the gradient of the loss function. At each iteration, MoE randomly samples a batch of tokens for training from $\mathcal{D}$ in an independent and identically distributed (i.i.d.) manner. Considering total T iterations, without loss of generality, let us define $\mathcal{B}^t \subseteq \mathcal{D}$ with size of B as the token batch at the t -th iteration, where $t \in \mathcal{T} = \{0, 1, \dots, T-1\}$ and $\mathcal{T}$ denotes the index set of iterations. Apparently, under sparse routing, the token batch routed to each expert is a subset of $\mathcal{B}^t$. In this context, let $\mathcal{X}_{(n,m)}^t \subseteq \mathcal{B}^t$ with size of $X_{(n,m)}^t$ denote the batch consisting of tokens routed to the (n, m) -th expert at the t -th iteration. Given $\mathbf{x}_i$ as an individual token in $\mathcal{B}^t$, let us define $\mathcal{A}_i^t = \bigcup_{n \in \mathcal{N}} \mathcal{A}_{i,n}^t$ as the index set of the activated expert for $\mathbf{x}_i$, where $\mathcal{A}_{i,n}^t$ denotes the index set of the activated experts for $\mathbf{x}_i$ in the n -th layer. Let $\mathbf{w}^t$ denote the parameter vector of the MoE-based model at the t -th iteration, the model parameters are updated as follows.

As detailed in Algorithm 1, the update of the parameters of each gate at the t -th iteration is given by

$$\begin{aligned} \mathbf{w}_{g,n}^{t+1} &= \mathbf{w}_{g,n}^t - \eta \nabla_{\mathbf{w}_{g,n}} F(\mathbf{w}^t, \mathcal{B}^t) \\ &= \mathbf{w}_{g,n}^t - \frac{1}{B} \sum_{\mathbf{x}_i \in \mathcal{B}_n^t} \eta \nabla_{\mathbf{w}_{g,n}} f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t), \forall n \in \mathcal{N}, \end{aligned} \quad (5)$$

where η denotes the learning rate. The update of the parameters of each expert at the t -th iteration is given by

$$\begin{aligned} \mathbf{w}_{(n,m)}^{t+1} &= \mathbf{w}_{(n,m)}^t - \eta \nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \mathcal{X}_{(n,m)}^t) \\ &= \mathbf{w}_{(n,m)}^t - \frac{1}{X_{(n,m)}^t} \sum_{\mathbf{x}_i \in \mathcal{X}_{(n,m)}^t} \eta \nabla_{\mathbf{w}_{(n,m)}} f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t), \\ &\forall (n, m) \in \mathcal{E}. \end{aligned} \quad (6)$$

Finally, the update of the non-MoE components at the t -th iteration is given by

$$\begin{aligned} \mathbf{w}_s^{t+1} &= \mathbf{w}_s^t - \eta \nabla_{\mathbf{w}_s} F(\mathbf{w}^t, \mathcal{B}^t) \\ &= \mathbf{w}_s^t - \frac{1}{B} \sum_{\mathbf{x}_i \in \mathcal{B}^t} \eta \nabla_{\mathbf{w}_s} f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t). \end{aligned} \quad (7)$$

4. Convergence Analysis

4.1. Motivations

Existing theoretical studies on MoE either focus on analyzing the performance gains over traditional dense models in specific scenarios (Chen et al., 2022; Chowdhury et al., 2023; Li et al., 2025; Li & Duan, 2025), or on establishing convergence guarantees for different routing mechanisms

Table 1. Comparison of MoE-based models between existing theoretical analysis and our work.

References	MoE Layer Number	Model Complexity	Model Parameter Scale	Impacts of Routing Sparsity
(Jordan & Xu, 1995)	Single	Low	Small	No
(Chen et al., 2022; Chowdhury et al., 2023)	Single	Medium	Small	No
(Nguyen et al., 2024a;b;c)	Single	Low	Small	No
(Li et al., 2025; Li & Duan, 2025)	Single	Low	Small	No
Our work	Multiple	High	Large	Yes

Algorithm 1 SGD for training MoE.

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1: Input: Total number of iterations  $T$ , number of activated experts per token  $K$ , batch size per iteration  $B$ , learning rate  $\eta$ , dataset  $\mathcal{D}$ ;
2: Initialization: Initialize model parameters  $\mathbf{w}^0$ ;
3: for  $t = 0$  to  $T - 1$  do
4:   Sample a batch  $\mathcal{B}^t \sim \mathcal{D}$ ;
5:   for Each  $\mathbf{x}_i \in \mathcal{B}^t$  in parallel do
6:     Forward Propagation:
7:     for  $n = 1$  to  $N$  do
8:       Process by non-MoE components;
9:       Generate routing scores  $g(\mathbf{w}_{g,n}^t, \mathbf{x}_i)$ ;
10:      Select top- $K$  experts and generates aggregated weights  $\pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{A}_{i,n}^t)$ ;
11:      for  $m = 1$  to  $M$  in parallel do
12:        if  $(n, m) \in \mathcal{A}_{i,n}^t$  then
13:          Route  $\mathbf{x}_i$  to the selected experts in  $\mathcal{A}_{i,n}^t$ ;
14:          Compute expert outputs  $E(\mathbf{w}_{(n,m)}^t, \mathbf{x}_i)$ .
15:        end if
16:        MoE  $(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{A}_{i,n}^t) \leftarrow \sum_{(n,m) \in \mathcal{A}_{i,n}^t} \pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{A}_{i,n}^t) [m] E(\mathbf{w}_{(n,m)}^t, \mathbf{x}_i)$ ;
17:      end for
18:    end for
19:    Compute loss  $f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t)$ ;
20:  end for
21:  Backward Propagation:
22:   $\mathbf{w}_{ne}^{t+1} \leftarrow \mathbf{w}_{ne}^t - \eta \nabla_{\mathbf{w}_{ne}} F(\mathbf{w}^t, \mathcal{B}^t)$ ,  $\mathbf{w}_{(n,m)}^{t+1} \leftarrow \mathbf{w}_{(n,m)}^t - \eta \nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \mathcal{X}_{(n,m)}^t)$ ;
23: end for
24: Output: Final model parameters  $\mathbf{w}^T$ .
    
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and expert architectures (Nguyen et al., 2024a;b;c), which largely overlook the impact of routing sparsity on convergence behavior of MoE training. This lack of theoretical understanding leaves MoE training that relies on heuristic or empirical choice of the number of activated experts per token prone to suboptimal performance. Moreover, to handle complex machine learning tasks, MoE-based models have evolved toward deeper architectures, higher model complexity, and substantially larger parameter scales (e.g., the applications of MoE in large language models (LLMs)). Existing theoretical analysis faces significant challenges in

generalizing to such modern MoE-based model training. Thus, motivated by these limitations, we provide one of the first theoretical analysis by using stochastic optimization theory to systematically analyze the impact of routing sparsity on MoE training, which is also applicable to modern deep and large-scale MoE-based models. Table 1 details the comparison between the MoE-based models adopted in prior theoretical studies and those addressed in this work.

4.2. Convergence Rate under Sparse Routing

In this subsection, we analyze the convergence rate of MoE training. From (5) to (7), unlike the gates and non-MoE components that need to process the entire token batch, i.e., $\mathcal{B}^t$, the experts only handle the batches routed to them. Based on this difference in the computing density, let us further categorize the components of the MoE-based model into expert parameters and non-expert parameters, where the vector of non-expert parameters at the t -th iteration is defined as

$$\mathbf{w}_{ne}^t = \text{concat}(\mathbf{w}_s^t, \mathbf{w}_{g,1}^t, \dots, \mathbf{w}_{g,N}^t). \quad (8)$$

In this context, we employ stochastic optimization theory to establish the convergence bound of MoE training. In what follows, we present several necessary assumptions to facilitate the subsequent analysis.

Assumption 1. For $\forall \mathbf{x}_i \in \mathcal{D}$, the loss function $f(\cdot)$ is non-convex with respect to the model parameter vector $\mathbf{w}$ (Foster et al., 2018).

Assumption 2. Let us define the set of activated experts for token $\mathbf{x}_i$ in the n -th layer as $\mathcal{A}_{i,n} \subseteq \mathcal{A}_i$. The loss function $f(\cdot)$ is ξ -Lipschitz (Liu et al., 2021a) with respect to the MoE output of a specific layer. Taking the n -th layer as an example, for $\forall \mathcal{A}_i$ and $\forall \mathcal{A}'_i = (\mathcal{A}_i \setminus \mathcal{A}_{i,n}) \cup \mathcal{A}'_{i,n}$, where $\mathcal{A}_{i,n} \neq \mathcal{A}'_{i,n}$, $f(\cdot)$ satisfies

$$\begin{aligned}
 & |f(\mathbf{w}, \mathbf{x}_i, \mathcal{A}'_i) - f(\mathbf{w}, \mathbf{x}_i, \mathcal{A}_i)| \\
 & \leq \xi \|\text{MoE}(\mathbf{w}_{g,n}, \mathbf{x}_i, \mathcal{A}'_{i,n}) - \text{MoE}(\mathbf{w}_{g,n}, \mathbf{x}_i, \mathcal{A}_{i,n})\| \\
 & = \xi \left\| \sum_{(n,m) \in \mathcal{A}'_{i,n}} \pi(\mathbf{w}_{g,n}, \mathbf{x}_i, \mathcal{A}'_{i,n}) [m] E(\mathbf{w}_{(n,m)}, \mathbf{x}_i) \right. \\
 & \quad \left. - \sum_{(n,m) \in \mathcal{A}_{i,n}} \pi(\mathbf{w}_{g,n}, \mathbf{x}_i, \mathcal{A}_{i,n}) [m] E(\mathbf{w}_{(n,m)}, \mathbf{x}_i) \right\|.
 \end{aligned} \quad (9)$$

Assumption 3. For $\forall x_i \in \mathcal{D}$, the spectral norms of the Hessian matrices of $f(\cdot)$ with respect to w_{ne} and $w_{(n,m)}$, $\forall (n,m) \in \mathcal{E}$ are upper bounded, i.e., there exist constants $l_1 > 0$, $l_2 > 0$, and $l_3 > 0$, such that the Hessian matrices satisfy

$$\|H_{w_{\text{ne}}, w_{\text{ne}}}\| = \left\| \frac{\partial^2 f(w, x_i, \mathcal{A}_i)}{\partial w_{\text{ne}}^2} \right\| \leq l_1, \quad (10)$$

$$\|H_{w_{(n,m)}, w_{\text{ne}}}\| = \left\| \frac{\partial^2 f(w, x_i, \mathcal{A}_i)}{\partial w_{(n,m)} \partial w_{\text{ne}}} \right\| \leq l_2, \quad (11)$$

$$\|H_{w_{(n,m)}, w_{(n',m')}}\| = \left\| \frac{\partial^2 f(w, x_i, \mathcal{A}_i)}{\partial w_{(n,m)} \partial w_{(n',m')}} \right\| \leq l_3, \quad (12)$$

$$\forall (n, m), (n', m') \in \mathcal{E}.$$

Assumption 4. For $\forall x_i$ sampled from $\mathcal{D}$ in an i.i.d. manner, the estimation of $\nabla_w f(\cdot)$ is unbiased (Li et al., 2019), i.e., $f(\cdot)$ satisfies

$$\mathbb{E}_{x_i} \{ \nabla_{w_{\text{ne}}} f(w, x_i, \mathcal{A}_i) \} = \nabla_{w_{\text{ne}}} F(w, \mathcal{D}), \quad (13)$$

and

$$\begin{aligned} \mathbb{E}_{x_i} \{ \nabla_{w_{(n,m)}} f(w, x_i, \mathcal{A}_i) | x_i \in \tilde{\mathcal{X}}_{(n,m)} \} \\ = \nabla_{w_{(n,m)}} F(w, \tilde{\mathcal{X}}_{(n,m)}), \quad \forall (n, m) \in \mathcal{E}, \end{aligned} \quad (14)$$

where $\tilde{\mathcal{X}}_{(n,m)} \subseteq \mathcal{D}$ denotes the subset of tokens routed to the (n, m) -th expert from the entire dataset $\mathcal{D}$ given w .

Assumption 5. For $\forall x_i$ sampled from $\mathcal{D}$ in an i.i.d. manner, the stochastic gradient grows with the magnitude of gradient expectation (Schmidt & Roux, 2013; Kawamoto & Iiduka, 2025), i.e., there exist constants $\beta > 0$, $\sigma > 0$, $\beta_{(n,m)} > 0$, and $\sigma_{(n,m)} > 0$, $\forall (n, m) \in \mathcal{E}$, such that $f(\cdot)$ satisfies

$$\mathbb{E}_{x_i} \{ \|\nabla_{w_{\text{ne}}} f(w, x_i, \mathcal{A}_i)\|^2 \} \leq \beta \|\nabla_{w_{\text{ne}}} F(w, \mathcal{D})\|^2 + \sigma^2, \quad (15)$$

and

$$\begin{aligned} \mathbb{E}_{x_i} \{ \|\nabla_{w_{(n,m)}} f(w, x_i, \mathcal{A}_i)\|^2 | x_i \in \tilde{\mathcal{X}}_{(n,m)} \} \\ \leq \beta_{(n,m)} \|\nabla_{w_{(n,m)}} F(w, \tilde{\mathcal{X}}_{(n,m)})\|^2 + \sigma_{(n,m)}^2, \\ \forall (n, m) \in \mathcal{E}. \end{aligned} \quad (16)$$

To analyze the convergence performance of MoE training, we evaluate the expected one-step decrease of the average loss function

$$\begin{aligned} \mathbb{E}_{\mathcal{B}^t} \{ F(w^{t+1}, \mathcal{D}) \} - F(w^t, \mathcal{D}) \\ = \frac{1}{D} \sum_{x_i \in \mathcal{D}} \mathbb{E}_{\mathcal{B}^t} \{ f(w^{t+1}, x_i, \mathcal{A}_i^{t+1}) \} - f(w^t, x_i, \mathcal{A}_i^t). \end{aligned} \quad (17)$$

Due to the randomness of the stochastic gradient algorithm and discrete top- K routing, the gates may activate different expert combinations for the same token (e.g., x_i) in the consecutive iterations if $K < M$, i.e., $\mathcal{A}_i^t \neq \mathcal{A}_i^{t+1}$. This results in the loss function $f(\cdot)$ being discontinuous. To address the analytical challenges posed by this discontinuity, we first analyze the gap of the average loss functions in the consecutive iterations, given all tokens are routed to the same set of experts. Specifically, we construct an auxiliary function as

$$\tilde{F}(w^{t+1}, \mathcal{D}) = \frac{1}{D} \sum_{x_i \in \mathcal{D}} f(w^{t+1}, x_i, \mathcal{A}_i^t), \quad (18)$$

and rewrite (17) as

$$\begin{aligned} \mathbb{E}_{\mathcal{B}^t} \{ F(w^{t+1}, \mathcal{D}) \} - F(w^t, \mathcal{D}) \\ = \mathbb{E}_{\mathcal{B}^t} \{ F(w^{t+1}, \mathcal{D}) \} - \mathbb{E}_{\mathcal{B}^t} \{ \tilde{F}(w^{t+1}, \mathcal{D}) \} \\ + \mathbb{E}_{\mathcal{B}^t} \{ \tilde{F}(w^{t+1}, \mathcal{D}) \} - F(w^t, \mathcal{D}) \end{aligned} \quad (19)$$

for further analysis, where we define $\mathbb{E}_{\mathcal{B}^t} \{ F(w^{t+1}, \mathcal{D}) \} - \mathbb{E}_{\mathcal{B}^t} \{ \tilde{F}(w^{t+1}, \mathcal{D}) \}$ as the routing-induced discrepancy.

With Assumptions 1-5, let us analyze the convergence performance of MoE training as follows.

Theorem 1. Assuming that the computation result of any expert satisfies $\|E(w_{(n,m)}^t, x_i)\| \leq \xi'$, $\forall (n, m) \in \mathcal{E}$, $\forall t \in \mathcal{T}$, the loss gap between the MoE-based model under sparse routing (i.e., top- K routing) and dense routing at the t -th iteration is upper bounded by

$$\begin{aligned} |f(w^t, x_i, \mathcal{E}) - f(w^t, x_i, \mathcal{A}_i^t)| \\ \leq 2\Xi \sum_{n \in \mathcal{N}} \left(1 - \sum_{(n,m) \in \mathcal{A}_{i,n}^t} \pi(w_{g,n}^t, x_i, \mathcal{E}_n) [m] \right) \\ = 2\Xi \sum_{n \in \mathcal{N}} \left(\sum_{(n,m) \in \mathcal{E}_n \setminus \mathcal{A}_{i,n}^t} \pi(w_{g,n}^t, x_i, \mathcal{E}_n) [m] \right), \end{aligned} \quad (20)$$

where $\Xi = \xi \xi'$ is a constant.

Proof. The proof of Theorem 1 is in Appendix A. $\square$

Remark 1. Theorem 1 indicates that the loss gap between the MoE-based model under sparse routing and dense routing is mainly controlled by the term $1 - \sum_{(n,m) \in \mathcal{A}_{i,n}^t} \pi(w_{g,n}^t, x_i, \mathcal{E}_n) [m]$, where $\sum_{(n,m) \in \mathcal{A}_{i,n}^t} \pi(w_{g,n}^t, x_i, \mathcal{E}_n) [m]$ denotes the sum of the top- K highest weights of the experts under dense routing. Consequently, the loss gap decreases as this sum approaches 1 and finally vanishes until $K = M$.

Corollary 1. Assume there exists $\delta^t, \forall t \in \mathcal{T}$ such that $\pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{E}_n)[m] \leq \frac{\delta^t}{M}$. The routing-induced discrepancy at the t -th iteration is upper bounded by

$$\mathbb{E}_{\mathcal{B}^t} \left\{ F(\mathbf{w}^t, \mathcal{D}) - \tilde{F}(\mathbf{w}^t, \mathcal{D}) \right\} \leq 4N\Xi\delta^t \left(1 - \frac{K}{M} \right). \quad (21)$$

Proof. The proof of Corollary 1 is in Appendix B. $\square$

Remark 2. Corollary 1 indicates that the routing-induced discrepancy is controlled by the number of activated experts per token K , i.e., the upper bound of the routing-induced discrepancy decreases as K increases. This insight implies that activating more experts per token leads to greater overlap between the sets of activated experts across consecutive iterations, thereby yielding a more stable routing path for each token. Moreover, this insight aligns with intuition: activating fewer experts per token increases the number of possible expert combinations, resulting in more variable routing paths and reduced overlap between the sets of activated experts in consecutive iterations. This variability can lead to unstable expert training and slow down the convergence rate of MoE.

Theorem 2. Given constants $\eta_0 > 0, 0 < \tilde{\mu}_1 \leq \tilde{\mu}_2 < 1$, and $0 < \mu_1 \leq \mu_2 < 1$ such that

$$\begin{aligned} \eta < \eta_0 < \min \left\{ \frac{B}{L_1(B-1+\beta)}, \frac{\|\nabla_{\mathbf{w}_{ne}} F(\mathbf{w}^t, \mathcal{D})\|^2 B}{L_1 \sigma^2} \right\}, \\ \min_{(n,m) \in \mathcal{E}, t \in \mathcal{T}} \left\{ \frac{X_{(n,m)}^t}{L_2 N K (X_{(n,m)}^t - 1 + \beta_{(n,m)})} \right\}, \\ \min_{(n,m) \in \mathcal{E}, t \in \mathcal{T}} \left\{ \frac{\|\nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \tilde{\mathcal{X}}_{(n,m)}^t)\|^2 X_{(n,m)}^t}{L_2 N K \sigma_{(n,m)}^2} \right\}, \end{aligned} \quad (22)$$

$\mathbb{E} \left\{ \frac{\tilde{X}_{(n,m)}^t}{D} \right\} \in [\tilde{\mu}_1, \tilde{\mu}_2]$, and $\mathbb{E} \left\{ \frac{X_{(n,m)}^t}{B} \right\} \in [\mu_1, \mu_2]$, the convergence bound of the average gradient norm of MoE over all iterations is given by

$$\begin{aligned} & \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \left\{ \|\nabla_{\mathbf{w}} F(\mathbf{w}^t, \mathcal{D})\|^2 \right\} \\ & \leq \underbrace{\frac{2(F(\mathbf{w}^0, \mathcal{D}) - F_{\text{inf}})}{\eta T} \left(1 + \frac{1}{\tilde{\mu}_1} \right)}_{A_1: \text{Initial bias}} \\ & + \underbrace{\frac{\eta L_1 \sigma^2}{B} + \frac{\sum_{(n,m) \in \mathcal{E}} \tilde{\mu}_2 \eta L_2 N K \sigma_{(n,m)}^2}{\tilde{\mu}_1 \mu_1 B}}_{A_2: \text{Gradient variances}} \\ & + \underbrace{\frac{8N\Xi\tilde{\delta}}{\eta T} \left(1 - \frac{K}{M} \right) \left(1 + \frac{1}{\tilde{\mu}_1} \right)}_{A_3: \text{Routing-induced discrepancy}}, \end{aligned} \quad (23)$$

where F_{inf} denotes the optimal value of (3), $L_1 = l_1 + l_2$ and $L_2 = l_2 + l_3$ are constants, and $\tilde{\delta}$ is also a constant that satisfies $\lim_{T \rightarrow \infty} \sum_{t=0}^{T-1} \delta^{t+1} \leq \tilde{\delta}$.

Proof. The proof of Theorem 2 is in Appendix C. $\square$

Remark 3. Theorem 2 indicates that the average gradient norm of MoE over all iterations is controlled by three main components: the initial bias (i.e., A_1), the gradient variances (i.e., A_2), and the routing-induced discrepancy (i.e., A_3). Specifically, A_1 depends on the initial model parameters $\mathbf{w}^0$, A_2 is determined by the bounds of stochastic gradients σ^2 and $\sigma_{(n,m)}^2$, and A_3 is controlled by the number of activated experts per token K . It is notable that, as the total number of T increases, the overall convergence upper bound decreases. However, different from A_1 and A_3 , A_2 remains non-vanishing as T increases. This implies that the average gradient norm of MoE over all iterations can only converge to the magnitude of A_2 . To further improve the convergence performance, we can properly decrease the learning rate η to reduce A_2 (see Corollary 2).

Corollary 2. If $\left\{ \frac{\tilde{X}_{(n,m)}^t}{D}, \frac{X_{(n,m)}^t}{B} \right\} = \Theta\left(\frac{K}{M}\right), \forall (n,m) \in \mathcal{E}, \forall t \in \mathcal{T}$ and let $\eta = \Theta\left(\frac{\eta_0}{\sqrt{T}}\right)$ hold, the asymptotic convergence upper bound of the average gradient norm of MoE over all iterations is given by

$$\begin{aligned} & \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \left\{ \|\nabla_{\mathbf{w}} F(\mathbf{w}^t, \mathcal{D})\|^2 \right\} \\ & \leq \mathcal{O} \left(\left(Z_1 + Z_3 \left(1 - \frac{K}{M} \right) \right) \frac{1 + \frac{M}{K}}{\sqrt{T}} \right) + \mathcal{O} \left(\frac{Z_2}{\sqrt{T}} \right), \end{aligned} \quad (24)$$

where $Z_1 = (\eta_0)^{-1} (F(\mathbf{w}^0, \mathcal{D}) - F_{\text{inf}})$ is related to the initial bias, $Z_2 = \eta_0 B^{-1} (MN \sum_{(n,m) \in \mathcal{E}} \sigma_{(n,m)}^2 + \sigma^2)$ is related to the gradient variances, and $Z_3 = (\eta_0)^{-1} N\Xi\tilde{\delta}$ is related to the routing-induced discrepancy.

Proof. The proof of Corollary 2 is in Appendix D. $\square$

Remark 4. Corollary 2 indicates that, by properly decreasing the learning rate η , the average gradient norm of MoE over all iterations converges to zero as the total number of iterations T increases. This observation demonstrates that the gradient of MoE-based model can converge to a stationary point at a rate of $\mathcal{O}\left(\frac{1}{\sqrt{T}}\right)$. Moreover, Corollary 2 captures the relationship between the average gradient norm of MoE over all iterations and the number of activated experts per token K . Specifically, the first term of the bound in (24) decreases as K increases and roughly scales as $\mathcal{O}\left(1 + \frac{M}{K}\right)$, which implies that activating more experts

per token can accelerate the convergence of MoE training. However, the second term of the bound in (24) is independent of K , revealing that increasing K does not influence the convergence of the stochastic gradient variance term.

Remark 5. Corollary 2 indicates that, while increasing K can accelerate MoE convergence, the marginal gain in convergence speed diminishes. This insight points out that, as K further increases, the convergence acceleration becomes less significant.

4.3. Performance-Cost Trade-off

In this subsection, let us analyze the trade-off between convergence performance and computational cost induced by routing sparsity.

Corollary 3. *Let us define r_g , r_e , and r_s (in FLOPs) as the per-token computational cost incurred by each gate, expert, and the non-MoE components, respectively. Since each token undergoes processing by a total of N gates, KN experts, and the non-MoE components, the per-token computational cost (in FLOPs) is given by*

$$r_0 = r_s + Nr_g + KNr_e. \quad (25)$$

Given the total computational budget R (in FLOPs) of MoE training, the average gradient norm of MoE over all iterations is upper bounded by

$$\begin{aligned} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \{ \|\nabla_{\mathbf{w}} F(\mathbf{w}^t, \mathcal{D})\|^2 \} &\leq \mathcal{O} \left(\frac{Z_2 \sqrt{Br_0}}{\sqrt{R}} \right) \\ &+ \mathcal{O} \left(\left(Z_1 + Z_3 \left(1 - \frac{K}{M} \right) \right) \frac{\left(1 + \frac{M}{K} \right) \sqrt{Br_0}}{\sqrt{R}} \right). \end{aligned} \quad (26)$$

Proof. The proof of Corollary 3 is in Appendix E. $\square$

Remark 6. From (25), it is observed that $r_0 = \mathcal{O}(K)$. Thus, under a fixed total computational budget R (in FLOPs), Corollary 3 proves that the convergence upper bound of MoE training scales as $\mathcal{O} \left(\sqrt{\frac{K}{R}} + \frac{M}{\sqrt{KR}} \right)$, which is convex with respect to K over the range $K \in [1, M]$. This insight implies the existence of an optimal $K^* \in [1, M]$ that achieves the best convergence performance given any R .

Corollary 4. *From Corollary 3, given a computational budget R (in FLOPs), the optimal K^* that minimizes the average gradient norm of MoE over all iterations satisfies*

$$1 \leq K^* \leq \min \left\{ \mathcal{O} \left(\frac{(Z_1 + Z_3) M}{Z_1 + Z_2 + Z_3} \right), M \right\}. \quad (27)$$

Proof. The proof of Corollary 4 is in Appendix F. $\square$

Remark 7. Corollary 4 indicates that the range of K^* is controlled by the ratio $\frac{Z_1 + Z_3}{Z_1 + Z_2 + Z_3}$. In particular, the value of

K^* tends to be small if $Z_2 \gg Z_1 + Z_3$ and large otherwise. Moreover, Corollary 4 proves that K^* is independent of computational cost R .

5. Experimental Results

In this section, we validate the theoretical analysis through experiments conducted on four NVIDIA RTX 5880ADA GPUs and two NVIDIA RTX 5090D GPUs.

5.1. Experiment Settings

(1) **Datasets:** We use 6 types of benchmark datasets, where the 2 types are image datasets for image classification: CIFAR-10 (Abouelnaga et al., 2016) and CIFAR-100 (Nguyen et al., 2021), and the other 4 types are text datasets for language generation: WikiText-103 (Merity et al., 2018), Alpaca (Cui et al., 2023), Dolly (Chen et al., 2024), and GSM8K (Cobbe et al., 2021).

(2) **Models:** We adopt different types of MoE-based models for different machine learning tasks: (I) For CIFAR-10 and CIFAR-100, we adopt single-layer MoE-based models, where the gate is a linear network and each expert is a multi-layer convolutional neural network (CNN). (II) For WikiText-103, Alpaca, Dolly, and GSM8K, we adopt decoder-only Transformer models with 8 Transformer blocks, where the Feed Forward Network (FFN) of each Transformer block is replaced with a MoE layer. Each MoE layer consists a gate and multiple experts, where all the gates are linear networks and all the experts are Multilayer Perceptrons (MLPs) with two layers.

(3) **Experiment parameters:** For CIFAR-10 and CIFAR-100, the batch size (the number of images) per iteration is set by $B = 256$, and the learning rates are set as follows: 1×10^{-3} for CIFAR-10 and 3×10^{-4} for CIFAR-100. For WikiText-103, Alpaca, Dolly, and GSM8K, the batch size (the number of tokens) per iteration is set to $B = 2048$, and the learning rate is set by 3×10^{-5} .

5.2. Loss Gap between Sparse and Dense Routing

As stated in Section 4.2, the loss function of MoE is discontinuous under sparse routing, thus, we introduce the routing-induced discrepancy term in convergence analysis. Since the routing-induced discrepancy term is corely derived from the loss gap of MoE between sparse routing and dense routing (see in (28)), we show the gap values on Alpaca, Dolly, and GSM8K vs T under different K in Fig 1, respectively.

It is observed that when $K > 1$, the loss gap decreases as K increases, and this trend holds for most iterations. Moreover, the loss gap gradually stabilizes as the total number of iterations increases. This observation is consistent with

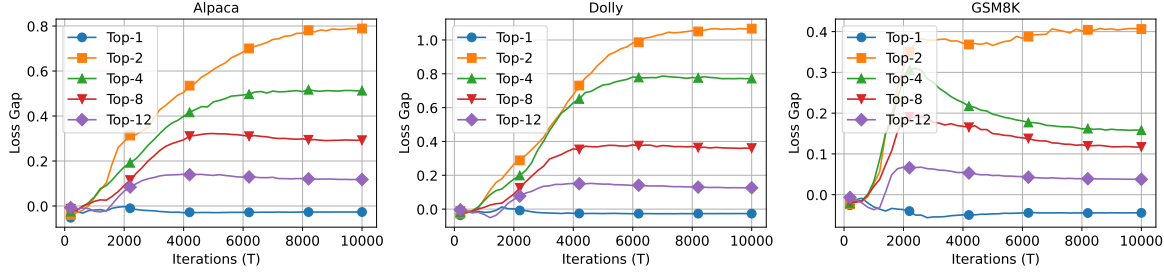


Figure 1. Loss gap vs T on Alpaca, Dolly, and GSM8K under different K , where the loss gap is the absolute difference between the loss values of sparse routing and dense routing.

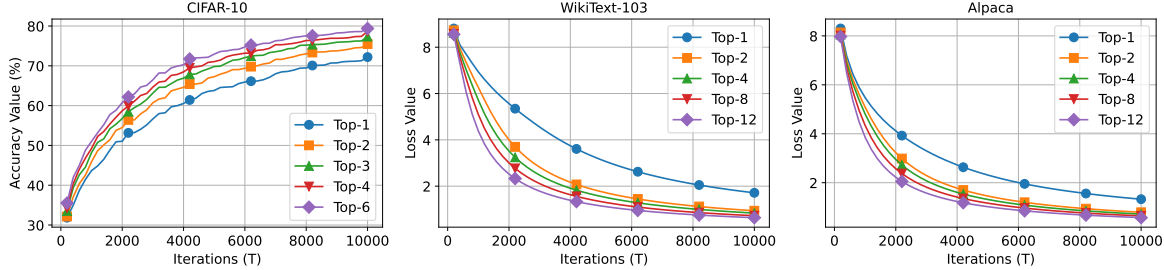


Figure 2. Loss (or Accuracy) values vs T on CIFAR-10, WikiText-103, and Alpaca under different K .

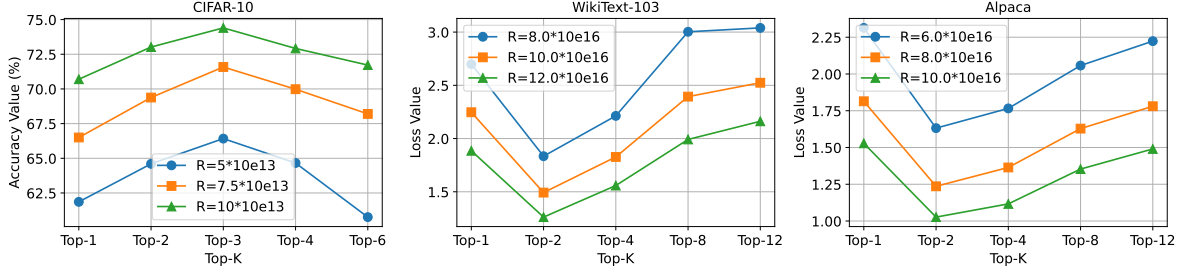


Figure 3. Loss (or Accuracy) values vs K on CIFAR-10, WikiText-103, and Alpaca under different R .

Theorem 1 and Corollary 1. However, it is also observed that the loss gap is the smallest when $K = 1$. This observation is not contradictory to our analytical results, as our results establish an upper bound on the loss gap and the gap under $K = 1$ does not exceed this theoretical bound. Moreover, the relatively small gap under $K = 1$ indicates that the routing choice of MoE is relatively invariant.

5.3. Convergence Rate under Sparse Routing

Fig. 2 illustrates the comparison of the convergence performance of MoE vs T under different K , wherein the benchmark datasets are CIFAR-10, WikiText-103, and Alpaca (the experimental results on the other datasets are shown in Appendix G). It is observed that MoE’s convergence speed accelerates as K increases. For instance, on WikiText-103, when converging to the same loss value (i.e., 1.71), the number of iterations required decreases from 10000 (under $K = 1$) to 3200 (under $K = 2$) (i.e., $3.125\times$ reduction of

iterations). However, it is also observed that the marginal gain in MoE’s convergence speed diminishes as K increases. For example, on WikiText-103, after the same number of iterations (e.g., 10000), when K increases from 1 to 2, the loss value decreases from 1.71 to 0.93, corresponding to a reduction of 0.78. In contrast, when K further increases from 2 to 4, the loss only decreases from 0.93 to 0.83, yielding a much smaller reduction of 0.10. These observations are consistent with the analysis in Corollary 2.

5.4. Convergence-Cost Trade-off

Fig. 3 illustrates the comparison of convergence performance vs K under the fixed total computational cost, wherein the benchmark datasets are CIFAR-10, WikiText-103, and Alpaca (the experimental results on the other datasets are shown in Appendix G). It is observed that under the same computational cost, the convergence performance of MoE first improves and then degrades as K increases,

showing a convex trend. This observation demonstrates the optimal $K^* \in [1, M]$ that achieves the best convergence performance under the fixed total computational cost, which is consistent with the analysis in Corollary 3. Moreover, an interesting fact is that the optimal K is much smaller than the total number of experts in a layer (e.g., $K^* = 2 \ll M = 16$ for WikiText-103 and Alpaca). The reason lies that: first, as K increases, the marginal gain in the convergence speed of MoE diminishes, causing the growth rate of computational cost to outpace the improvement in convergence speed (as analyzed in Remarks 5 and 6); second, due to the massive training data and large-scale of MoE-based models in terms of parameter size and the number of experts, the magnitude of Z_2 in (24) tends to be significantly larger than that of $Z_1 + Z_3$ in (24), which leads to a small K^* (i.e., $K^* \ll M$) as analyzed in Remark 7. This observation is consistent with the empirical implementation of MoE in (Lepikhin et al., 2021; Fedus et al., 2022; Wei et al., 2024), wherein K is set by 1 or 2 (i.e., $\ll M$) to achieve better trade-off between computational cost and convergence performance.

6. Conclusion

In this work, we provided one of the first convergence upper bound of MoE training that revealed the relationships among routing sparsity, convergence behavior, and computational cost. Given sufficient computational budget, we proved the convergence upper bound scaled as $\mathcal{O}\left(\frac{1+M/K}{\sqrt{T}}\right)$, which not only theoretically guaranteed the convergence of MoE training, but also indicated that activating more experts per token (i.e., enlarging K) could accelerate convergence. Moreover, under a fixed computational budget R (in FLOPs), we refined the convergence upper bound that scaled as $\mathcal{O}\left(\sqrt{\frac{K}{R}} + \frac{M}{\sqrt{KR}}\right)$, which was convex with respect to K over the range $K \in [1, M]$. This insight demonstrated the existence of $K^* \in [1, M]$ that achieved the best convergence performance given any R . Finally, extensive experimental results under diverse settings were conducted to corroborate our theoretical findings. While the current experiments validate the existence of K^* under fixed- M settings, future work will extend the analysis to an arbitrary number of experts M to further capture the scaling relationship between K^* and M .

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Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Proof of Theorem 1

Theorem 1. Assuming that the computation result of any expert satisfies $\|E(\mathbf{w}_{(n,m)}^t, \mathbf{x}_i)\| \leq \xi'$, $\forall (n, m) \in \mathcal{E}$, the gap between the loss functions of the MoE with top- K routing and that with dense routing at the t -th iteration is upper bounded by

$$\begin{aligned} |f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{E}) - f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t)| &\leq 2\Xi \sum_{n \in \mathcal{N}} \left(1 - \sum_{(n,m) \in \mathcal{A}_{i,n}^t} \pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{E}_n)[m] \right) \\ &= 2\Xi \sum_{n \in \mathcal{N}} \left(\sum_{(n,m) \in \mathcal{E}_n \setminus \mathcal{A}_{i,n}^t} \pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{E}_n)[m] \right), \end{aligned} \quad (28)$$

where $\Xi = \xi\xi'$ is a constant.

Proof. We define an auxiliary activated expert set $\mathcal{A}_{\text{aux},i,n}^t$ for token $\mathbf{x}_i$ at iteration t as follows:

$$\mathcal{A}_{\text{aux},i,n}^t = \begin{cases} \mathcal{A}_{\text{aux},i,n+1}^t \setminus \mathcal{E}_n \cup \mathcal{A}_{i,n}^t, & \text{if } n \neq N, \\ \mathcal{E} \setminus \mathcal{E}_N \cup \mathcal{A}_{i,N}^t, & \text{if } n = N. \end{cases} \quad (29)$$

With Assumption 2, we have

$$\begin{aligned} &\left| f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{E}) - f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t) \right| \\ &= \left| f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{E}) - f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_{\text{aux},i,N}^t) + \sum_{n=1}^{N-1} (f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_{\text{aux},i,n+1}^t) - f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_{\text{aux},i,n}^t)) \right| \\ &\leq \left| f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{E}) - f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_{\text{aux},i,N}^t) \right| + \sum_{n=1}^{N-1} \left| f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_{\text{aux},i,n+1}^t) - f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_{\text{aux},i,n}^t) \right|. \end{aligned} \quad (30)$$

For $|f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_{\text{aux},i,n+1}^t) - f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_{\text{aux},i,n}^t)|$, we have

$$\begin{aligned} &\left| f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_{\text{aux},i,n+1}^t) - f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_{\text{aux},i,n}^t) \right| \\ &\leq \xi \left\| \sum_{(n,m) \in \mathcal{E}} \left(\pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{E}_n)[m] E(\mathbf{w}_{(n,m)}^t, \mathbf{x}_i) - \pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{A}_{i,n}^t)[m] E(\mathbf{w}_{(n,m)}^t, \mathbf{x}_i) \right) \right\|. \end{aligned} \quad (31)$$

If $\|E(\mathbf{w}_{(n,m)}^t, \mathbf{x}_i)\| \leq \xi'$, we have

$$\begin{aligned} &\xi \left\| \sum_{(n,m) \in \mathcal{E}} \left(\pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{E}_n)[m] E(\mathbf{w}_{(n,m)}^t, \mathbf{x}_i) - \pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{A}_{i,n}^t)[m] E(\mathbf{w}_{(n,m)}^t, \mathbf{x}_i) \right) \right\| \\ &\leq \xi\xi' \sum_{(n,m) \in \mathcal{E}} \left| \pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{E}_n)[m] - \pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{A}_{i,n}^t)[m] \right|. \end{aligned} \quad (32)$$

Since $\pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{E}_n)[m] = \frac{\exp(g(\mathbf{w}_{g,n}^t, \mathbf{x}_i)[m])}{\sum_{(n,m') \in \mathcal{E}_n} \exp(g(\mathbf{w}_{g,n}^t, \mathbf{x}_i)[m'])}$ and $\pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{A}_{i,n}^t)[m] = \frac{\exp(g(\mathbf{w}_{g,n}^t, \mathbf{x}_i)[m])}{\sum_{(n,m') \in \mathcal{A}_{i,n}^t} \exp(g(\mathbf{w}_{g,n}^t, \mathbf{x}_i)[m'])}$, if $(n, m) \in \mathcal{A}_{i,n}^t$ we have

$$\left| \pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{E}_n)[m] - \pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{A}_{i,n}^t)[m] \right|$$

$$\begin{aligned}
 &= \left| \frac{\exp(g(\mathbf{w}_{g,n}^t, \mathbf{x}_i)[m])}{\sum_{(n,m') \in \mathcal{E}_n} \exp(g(\mathbf{w}_{g,n}^t, \mathbf{x}_i)[m'])} - \frac{\exp(g(\mathbf{w}_{g,n}^t, \mathbf{x}_i)[m])}{\sum_{(n,m') \in \mathcal{A}_{i,n}^t} \exp(g(\mathbf{w}_{g,n}^t, \mathbf{x}_i)[m'])} \right| \\
 &= \exp(g(\mathbf{w}_{g,n}^t, \mathbf{x}_i)[m]) \left| \frac{\sum_{(n,m') \in \mathcal{A}_{i,n}^t} \exp(g(\mathbf{w}_{g,n}^t, \mathbf{x}_i)[m']) - \sum_{(n,m') \in \mathcal{E}_n} \exp(g(\mathbf{w}_{g,n}^t, \mathbf{x}_i)[m'])}{\sum_{(n,m') \in \mathcal{E}_n} \exp(g(\mathbf{w}_{g,n}^t, \mathbf{x}_i)[m']) \sum_{(n,m') \in \mathcal{A}_{i,n}^t} \exp(g(\mathbf{w}_{g,n}^t, \mathbf{x}_i)[m'])} \right| \\
 &= \pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{A}_{i,n}^t)[m] \left| \frac{\sum_{(n,m') \in \mathcal{E}_n \setminus \mathcal{A}_{i,n}^t} \exp(g(\mathbf{w}_{g,n}^t, \mathbf{x}_i)[m'])}{\sum_{(n,m') \in \mathcal{E}_n} \exp(g(\mathbf{w}_{g,n}^t, \mathbf{x}_i)[m'])} \right| \\
 &= \pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{A}_{i,n}^t)[m] \sum_{(n,m') \in \mathcal{E}_n \setminus \mathcal{A}_{i,n}^t} \pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{E}_n)[m'] \\
 &= \pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{A}_{i,n}^t)[m] \left(1 - \sum_{(n,m') \in \mathcal{A}_{i,n}^t} \pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{E}_n)[m'] \right), \forall (n, m) \in \mathcal{A}_{i,n}^t. \tag{33}
 \end{aligned}$$

In this context, we further have

$$\begin{aligned}
 &\sum_{(n,m) \in \mathcal{E}_n} \left| \pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{E}_n)[m] - \pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{A}_{i,n}^t)[m] \right| \\
 &= \sum_{(n,m) \in \mathcal{A}_{i,n}^t} \left| \pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{E}_n)[m] - \pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{A}_{i,n}^t)[m] \right| + \sum_{(n,m) \in \mathcal{E}_n \setminus \mathcal{A}_{i,n}^t} \pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{E}_n)[m] \\
 &= \sum_{(n,m) \in \mathcal{A}_{i,n}^t} \pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{A}_{i,n}^t)[m] \left(1 - \sum_{(n,m') \in \mathcal{A}_{i,n}^t} \pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{E}_n)[m'] \right) + \sum_{(n,m) \in \mathcal{E}_n \setminus \mathcal{A}_{i,n}^t} \pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{E}_n)[m] \\
 &= 2 \sum_{(n,m) \in \mathcal{E}_n \setminus \mathcal{A}_{i,n}^t} \pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{E}_n)[m]. \tag{34}
 \end{aligned}$$

Finally, we have

$$\left| f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{E}) - f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t) \right| \leq 2\Xi \sum_{n \in \mathcal{N}} \left(1 - \sum_{(n,m) \in \mathcal{A}_{i,n}^t} \pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{E}_n)[m] \right), \tag{35}$$

where $\Xi = \xi \xi'$. Thus, we complete the proof. $\square$

B. Proof of Corollary 1

Corollary 1. *If there exist δ^t , $\forall t \in \mathcal{T}$ such that $\pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{E}_n)[m] \leq \frac{\delta^t}{M}$, the routing-induced discrepancy at the t -th iteration is upper bounded by*

$$\mathbb{E}_{\mathcal{B}^t} \{F(\mathbf{w}^t, \mathcal{D})\} - \mathbb{E}_{\mathcal{B}^t} \{\tilde{F}(\mathbf{w}^t, \mathcal{D})\} \leq 4N\Xi\delta^t \left(1 - \frac{K}{M} \right). \tag{36}$$

Proof. Since

$$\begin{aligned}
 &\mathbb{E}_{\mathcal{B}^t} \{F(\mathbf{w}^t, \mathcal{D})\} - \mathbb{E}_{\mathcal{B}^t} \{\tilde{F}(\mathbf{w}^t, \mathcal{D})\} \leq \frac{1}{D} \sum_{\mathbf{x}_i \in \mathcal{D}} \mathbb{E}_{\mathcal{B}^t} \{ |f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t) - f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^{t-1})| \} \\
 &= \frac{1}{D} \sum_{\mathbf{x}_i \in \mathcal{D}} \mathbb{E}_{\mathcal{B}^t} \left\{ \left| f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{E}) - f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t) + f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{E}) - f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^{t-1}) \right| \right\} \\
 &\leq \frac{1}{D} \sum_{\mathbf{x}_i \in \mathcal{D}} \mathbb{E}_{\mathcal{B}^t} \left\{ \left| f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{E}) - f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t) \right| + \left| f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{E}) - f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^{t-1}) \right| \right\}
 \end{aligned}$$

$$\leq \frac{2\Xi}{D} \sum_{\mathbf{x}_i \in \mathcal{D}} \mathbb{E}_{\mathcal{B}^t} \left\{ \sum_{n \in \mathcal{N}} \left(1 - \sum_{(n,m) \in \mathcal{A}_{i,n}^t} \pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{E}_n) [m] \right) + \sum_{n \in \mathcal{N}} \left(1 - \sum_{(n,m) \in \mathcal{A}_{i,n}^{t-1}} \pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{E}_n) [m] \right) \right\}, \quad (37)$$

if there exists $\delta^t > 0$ ($\forall t \in \mathcal{T}$) such that $\pi(\mathbf{w}_{g,n}^t, \mathbf{x}_i, \mathcal{E}_n) \geq \frac{\delta^t}{M}$, we have

$$\mathbb{E}_{\mathcal{B}^t} \{F(\mathbf{w}^t, \mathcal{D})\} - \mathbb{E}_{\mathcal{B}^t} \{\tilde{F}(\mathbf{w}^t, \mathcal{D})\} \leq \frac{4\Xi N (M - K) \delta^t}{M}. \quad (38)$$

Thus, we complete the proof. $\square$

C. Proof of Theorem 2

Theorem 2. Given constants $\eta_0 > 0$, $0 < \tilde{\mu}_1 \leq \tilde{\mu}_2 < 1$, and $0 < \mu_1 \leq \mu_2 < 1$ such that

$$\begin{aligned} \eta < \eta_0 < \min & \left\{ \frac{B}{L_1 (B - 1 + \beta)}, \frac{\|\nabla_{\mathbf{w}_{ne}} F(\mathbf{w}^t, \mathcal{D})\|^2 B}{L_1 \sigma^2}, \right. \\ & \min_{(n,m) \in \mathcal{E}, t \in \mathcal{T}} \left\{ \frac{X_{(n,m)}^t}{L_2 N K (X_{(n,m)}^t - 1 + \beta_{(n,m)})} \right\}, \\ & \min_{(n,m) \in \mathcal{E}, t \in \mathcal{T}} \left\{ \frac{\|\nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \tilde{\mathcal{X}}_{(n,m)}^t)\|^2 X_{(n,m)}^t}{L_2 N K \sigma_{(n,m)}^2} \right\} \Big\}, \end{aligned} \quad (39)$$

$\mathbb{E} \left\{ \frac{\tilde{X}_{(n,m)}^t}{D} \right\} \in [\tilde{\mu}_1, \tilde{\mu}_2]$, and $\mathbb{E} \left\{ \frac{X_{(n,m)}^t}{B} \right\} \in [\mu_1, \mu_2]$, the convergence bound of the average gradient norm of MoE over all iterations is given by

$$\begin{aligned} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \{ \|\nabla_{\mathbf{w}} F(\mathbf{w}^t, \mathcal{D})\|^2 \} & \leq \underbrace{\frac{2(F(\mathbf{w}^0, \mathcal{D}) - F_{\inf})}{\eta T} \left(1 + \frac{1}{\tilde{\mu}_1} \right)}_{A_1: \text{Initial bias}} \\ & + \underbrace{\frac{\eta L_1 \sigma^2}{B} + \frac{\sum_{(n,m) \in \mathcal{E}} \tilde{\mu}_2 \eta L_2 N K \sigma_{(n,m)}^2}{\tilde{\mu}_1 \mu_1 B}}_{A_2: \text{Gradient variances}} + \underbrace{\frac{8N\Xi\tilde{\delta}}{\eta T} \left(1 - \frac{K}{M} \right) \left(1 + \frac{1}{\tilde{\mu}_1} \right)}_{A_3: \text{Routing-induced discrepancy}}, \end{aligned} \quad (40)$$

where $F_{\inf}$ denotes the optimal value of (3), $L_1 = l_1 + l_2$ and $L_2 = l_2 + l_3$ are constants, and $\tilde{\delta}$ is also a constant that satisfies $\lim_{T \rightarrow \infty} \sum_{t=0}^{T-1} \delta^{t+1} \leq \tilde{\delta}$.

Proof. For $\forall \mathbf{x}_i \in \mathcal{D}$, the second-order Taylor expansion of $f(\cdot)$ is given by

$$f(\mathbf{w}^{t+1}, \mathbf{x}_i, \mathcal{A}_i^t) \approx f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t) + \langle \nabla_{\mathbf{w}} f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t), \mathbf{w}^{t+1} - \mathbf{w}^t \rangle + \frac{1}{2} (\mathbf{w}^{t+1} - \mathbf{w}^t)^\top \mathbf{H}(f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t)) (\mathbf{w}^{t+1} - \mathbf{w}^t), \quad (41)$$

where $\mathbf{H}(\cdot)$ denotes the Hessian matrix with respect to $f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t)$. Specifically, $\mathbf{H}(\cdot)$ can be expressed in a block form as

$$\begin{pmatrix} \mathbf{H}_{\mathbf{w}_{ne}, \mathbf{w}_{ne}} & \mathbf{H}_{\mathbf{w}_{ne}, \mathbf{w}_{(1,1)}} & \cdots & \mathbf{H}_{\mathbf{w}_{ne}, \mathbf{w}_{(N,M)}} \\ \mathbf{H}_{\mathbf{w}_{(1,1)}, \mathbf{w}_{ne}} & \mathbf{H}_{\mathbf{w}_{(1,1)}, \mathbf{w}_{(1,1)}} & \cdots & \mathbf{H}_{\mathbf{w}_{(1,1)}, \mathbf{w}_{(N,M)}} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{H}_{\mathbf{w}_{(N,M)}, \mathbf{w}_{ne}} & \mathbf{H}_{\mathbf{w}_{(N,M)}, \mathbf{w}_{(1,1)}} & \cdots & \mathbf{H}_{\mathbf{w}_{(N,M)}, \mathbf{w}_{(N,M)}} \end{pmatrix}, \quad (42)$$

where $\mathbf{H}_{\mathbf{w}, \mathbf{w}'} = \frac{\partial^2 f}{\partial \mathbf{w} \partial \mathbf{w}'}$. Thus, (41) can be rewrite as

$$\begin{aligned}
 f(\mathbf{w}^{t+1}, \mathbf{x}_i, \mathcal{A}_i^t) - f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t) &\approx \langle \nabla_{\mathbf{w}_{\text{ne}}} f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t), \mathbf{w}_{\text{ne}}^{t+1} - \mathbf{w}_{\text{ne}}^t \rangle \\
 &+ \sum_{(n,m) \in \mathcal{E}} \langle \nabla_{\mathbf{w}_{(n,m)}} f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t), \mathbf{w}_{(n,m)}^{t+1} - \mathbf{w}_{(n,m)}^t \rangle + \frac{1}{2} (\mathbf{w}_{\text{ne}}^{t+1} - \mathbf{w}_{\text{ne}}^t)^T \mathbf{H}_{\mathbf{w}_{\text{ne}}, \mathbf{w}_{\text{ne}}} (\mathbf{w}_{\text{ne}}^{t+1} - \mathbf{w}_{\text{ne}}^t) \\
 &+ \frac{1}{2} (\mathbf{w}_{\text{ne}}^{t+1} - \mathbf{w}_{\text{ne}}^t)^T \sum_{(n,m) \in \mathcal{E}} \mathbf{H}_{\mathbf{w}_{\text{ne}}, \mathbf{w}_{(n,m)}} (\mathbf{w}_{(n,m)}^{t+1} - \mathbf{w}_{(n,m)}^t) \\
 &+ \frac{1}{2} \sum_{(n,m) \in \mathcal{E}} \sum_{(n',m') \in \mathcal{E}} (\mathbf{w}_{(n,m)}^{t+1} - \mathbf{w}_{(n,m)}^t)^T \mathbf{H}_{\mathbf{w}_{(n,m)}, \mathbf{w}_{(n',m')}} (\mathbf{w}_{(n',m')}^{t+1} - \mathbf{w}_{(n',m')}^t). \tag{43}
 \end{aligned}$$

With top- K routing, if $(n, m) \notin \mathcal{A}_i^t$, we have $\nabla_{\mathbf{w}_{(n,m)}} f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t) = \mathbf{0}$ and $\mathbf{H}_{\mathbf{w}_{(n,m)}, \mathbf{w}_{(n',m')}} = \mathbf{H}_{\mathbf{w}_{(n',m')}, \mathbf{w}_{(n,m)}} = \mathbf{0}$. In this context, (43) can be rewrite as

$$\begin{aligned}
 f(\mathbf{w}^{t+1}, \mathbf{x}_i, \mathcal{A}_i^t) - f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t) &\approx \langle \nabla_{\mathbf{w}_{\text{ne}}} f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t), \mathbf{w}_{\text{ne}}^{t+1} - \mathbf{w}_{\text{ne}}^t \rangle \\
 &+ \sum_{(n,m) \in \mathcal{A}_i^t} \langle \nabla_{\mathbf{w}_{(n,m)}} f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t), \mathbf{w}_{(n,m)}^{t+1} - \mathbf{w}_{(n,m)}^t \rangle + \frac{1}{2} (\mathbf{w}_{\text{ne}}^{t+1} - \mathbf{w}_{\text{ne}}^t)^T \mathbf{H}_{\mathbf{w}_{\text{ne}}, \mathbf{w}_{\text{ne}}} (\mathbf{w}_{\text{ne}}^{t+1} - \mathbf{w}_{\text{ne}}^t) \\
 &+ \frac{1}{2} (\mathbf{w}_{\text{ne}}^{t+1} - \mathbf{w}_{\text{ne}}^t)^T \sum_{(n,m) \in \mathcal{A}_i^t} \mathbf{H}_{\mathbf{w}_{\text{ne}}, \mathbf{w}_{(n,m)}} (\mathbf{w}_{(n,m)}^{t+1} - \mathbf{w}_{(n,m)}^t) \\
 &+ \frac{1}{2} \sum_{(n,m) \in \mathcal{A}_i^t} (\mathbf{w}_{(n,m)}^{t+1} - \mathbf{w}_{(n,m)}^t)^T \mathbf{H}_{\mathbf{w}_{\text{ne}}, \mathbf{w}_{(n,m)}} (\mathbf{w}_{\text{ne}}^{t+1} - \mathbf{w}_{\text{ne}}^t) \\
 &+ \frac{1}{2} \sum_{(n,m) \in \mathcal{A}_i^t} \sum_{(n',m') \in \mathcal{A}_i^t} (\mathbf{w}_{(n,m)}^{t+1} - \mathbf{w}_{(n,m)}^t)^T \mathbf{H}_{\mathbf{w}_{(n,m)}, \mathbf{w}_{(n',m')}} (\mathbf{w}_{(n',m')}^{t+1} - \mathbf{w}_{(n',m')}^t). \tag{44}
 \end{aligned}$$

Since $\mathbf{x}^T \mathbf{H} \mathbf{y} \leq \|\mathbf{x}\| \|\mathbf{H}\| \|\mathbf{y}\|$ and if $\|\mathbf{H}_{\mathbf{w}_{\text{ne}}, \mathbf{w}_{\text{ne}}}\| \leq l_1$, $\|\mathbf{H}_{\mathbf{w}_{\text{ne}}, \mathbf{w}_{(n,m)}}\| = \|\mathbf{H}_{\mathbf{w}_{(n,m)}, \mathbf{w}_{\text{ne}}}\| \leq l_2$, and $\|\mathbf{H}_{\mathbf{w}_{(n,m)}, \mathbf{w}_{(n',m')}}\| = \|\mathbf{H}_{\mathbf{w}_{(n',m')}, \mathbf{w}_{(n,m)}}\| \leq l_3$ ($\forall \mathbf{x}_i \in \mathcal{D}, (n, m), (n', m') \in \mathcal{E}, t \in \mathcal{T}$), (44) is upper bounded by

$$\begin{aligned}
 f(\mathbf{w}^{t+1}, \mathbf{x}_i, \mathcal{A}_i^t) - f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t) &\leq \langle \nabla_{\mathbf{w}_{\text{ne}}} f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t), \mathbf{w}_{\text{ne}}^{t+1} - \mathbf{w}_{\text{ne}}^t \rangle \\
 &+ \sum_{(n,m) \in \mathcal{A}_i^t} \langle \nabla_{\mathbf{w}_{(n,m)}} f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t), \mathbf{w}_{(n,m)}^{t+1} - \mathbf{w}_{(n,m)}^t \rangle + \frac{l_1}{2} \|\mathbf{w}_{\text{ne}}^{t+1} - \mathbf{w}_{\text{ne}}^t\|^2 \\
 &+ l_2 \|\mathbf{w}_{\text{ne}}^{t+1} - \mathbf{w}_{\text{ne}}^t\| \sum_{(n,m) \in \mathcal{A}_i^t} \|\mathbf{w}_{(n,m)}^{t+1} - \mathbf{w}_{(n,m)}^t\| + \frac{l_3}{2} \sum_{(n,m) \in \mathcal{A}_i^t} \sum_{(n',m') \in \mathcal{A}_i^t} \|\mathbf{w}_{(n,m)}^{t+1} - \mathbf{w}_{(n,m)}^t\| \|\mathbf{w}_{(n',m')}^{t+1} - \mathbf{w}_{(n',m')}^t\|. \tag{45}
 \end{aligned}$$

Since $\sum_{x \in \mathcal{N}} \sum_{y \in \mathcal{N}} f(x) f(y) = (\sum_{x \in \mathcal{N}} f(x))^2 \leq N \sum_{x \in \mathcal{N}} (f(x))^2$ and $xy \leq \frac{1}{2}x^2 + \frac{1}{2}y^2$ we have

$$\begin{aligned}
 \|\mathbf{w}_{\text{ne}}^{t+1} - \mathbf{w}_{\text{ne}}^t\| \sum_{(n,m) \in \mathcal{A}_i^t} \|\mathbf{w}_{(n,m)}^{t+1} - \mathbf{w}_{(n,m)}^t\| &\leq \frac{1}{2} \|\mathbf{w}_{\text{ne}}^{t+1} - \mathbf{w}_{\text{ne}}^t\|^2 + \frac{1}{2} \left(\sum_{(n,m) \in \mathcal{A}_i^t} \|\mathbf{w}_{(n,m)}^{t+1} - \mathbf{w}_{(n,m)}^t\| \right)^2 \\
 &\leq \frac{1}{2} \|\mathbf{w}_{\text{ne}}^{t+1} - \mathbf{w}_{\text{ne}}^t\|^2 + \frac{NK}{2} \sum_{(n,m) \in \mathcal{A}_i^t} \|\mathbf{w}_{(n,m)}^{t+1} - \mathbf{w}_{(n,m)}^t\|^2 \tag{46}
 \end{aligned}$$

and

$$\begin{aligned}
 \sum_{(n,m) \in \mathcal{A}_i^t} \sum_{(n',m') \in \mathcal{A}_i^t} \|\mathbf{w}_{(n,m)}^{t+1} - \mathbf{w}_{(n,m)}^t\| \|\mathbf{w}_{(n',m')}^{t+1} - \mathbf{w}_{(n',m')}^t\| &\leq \left(\sum_{(n,m) \in \mathcal{A}_i^t} \|\mathbf{w}_{(n,m)}^{t+1} - \mathbf{w}_{(n,m)}^t\| \right)^2 \\
 &\leq NK \sum_{(n,m) \in \mathcal{A}_i^t} \|\mathbf{w}_{(n,m)}^{t+1} - \mathbf{w}_{(n,m)}^t\|^2. \tag{47}
 \end{aligned}$$

Thus, we further have

$$\begin{aligned}
 f(\mathbf{w}^{t+1}, \mathbf{x}_i, \mathcal{A}_i^t) - f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t) &\leq \langle \nabla_{\mathbf{w}_{\text{nc}}} f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t), \mathbf{w}_{\text{nc}}^{t+1} - \mathbf{w}_{\text{nc}}^t \rangle \\
 &+ \sum_{(n,m) \in \mathcal{A}_i^t} \langle \nabla_{\mathbf{w}_{(n,m)}} f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t), \mathbf{w}_{(n,m)}^{t+1} - \mathbf{w}_{(n,m)}^t \rangle + \frac{L_1}{2} \|\mathbf{w}_{\text{nc}}^{t+1} - \mathbf{w}_{\text{nc}}^t\|^2 \\
 &+ \frac{L_2 NK}{2} \sum_{(n,m) \in \mathcal{A}_i^t} \|\mathbf{w}_{(n,m)}^{t+1} - \mathbf{w}_{(n,m)}^t\|^2,
 \end{aligned} \tag{48}$$

where $L_1 = l_1 + l_2$ and $L_2 = l_2 + l_3$.

Since

$$\mathbb{E}_{\mathcal{B}^t} \left\{ \tilde{F}(\mathbf{w}^{t+1}, \mathcal{D}) \right\} - F(\mathbf{w}^t, \mathcal{D}) = \mathbb{E}_{\mathcal{B}^t} \left\{ \frac{1}{D} \left(\sum_{\mathbf{x}_i \in \mathcal{D}} f(\mathbf{w}^{t+1}, \mathbf{x}_i, \mathcal{A}_i^t) - f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t) \right) \right\}, \tag{49}$$

with Assumption 3, we have

$$\begin{aligned}
 \mathbb{E}_{\mathcal{B}^t} \left\{ \tilde{F}(\mathbf{w}^{t+1}, \mathcal{D}) \right\} - F(\mathbf{w}^t, \mathcal{D}) &= \mathbb{E}_{\mathcal{B}^t} \left\{ \frac{1}{D} \sum_{\mathbf{x}_i \in \mathcal{D}} (f(\mathbf{w}^{t+1}, \mathbf{x}_i, \mathcal{A}_i^t) - f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t)) \right\} \\
 &\leq \underbrace{\frac{1}{D} \sum_{\mathbf{x}_i \in \mathcal{D}} \left\langle \nabla_{\mathbf{w}_{\text{nc}}} f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t), \mathbb{E}_{\mathcal{B}^t} \left\{ \mathbf{w}_{\text{nc}}^{t+1} - \mathbf{w}_{\text{nc}}^t \right\} \right\rangle}_{\text{I}} + \underbrace{\frac{L_1}{2} \mathbb{E}_{\mathcal{B}^t} \left\{ \left\| \mathbf{w}_{\text{nc}}^{t+1} - \mathbf{w}_{\text{nc}}^t \right\|^2 \right\}}_{\text{II}} \\
 &+ \underbrace{\frac{1}{D} \sum_{\mathbf{x}_i \in \mathcal{D}} \sum_{(n,m) \in \mathcal{E}} \mathbf{1}((n,m) \in \mathcal{A}_i^t) \left(\left\langle \nabla_{\mathbf{w}_{(n,m)}} f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t), \mathbb{E}_{\mathcal{X}_{(n,m)}^t} \left\{ \mathbf{w}_{(n,m)}^{t+1} - \mathbf{w}_{(n,m)}^t \mid \mathcal{X}_{(n,m)}^t \subseteq \mathcal{B}^t \right\} \right\rangle \right)}_{\text{III}} \\
 &+ \underbrace{\sum_{(n,m) \in \mathcal{E}} \mathbf{1}((n,m) \in \mathcal{A}_i^t) \left(\frac{L_2 NK}{2} \mathbb{E}_{\mathcal{X}_{(n,m)}^t} \left\{ \left\| \mathbf{w}_{(n,m)}^{t+1} - \mathbf{w}_{(n,m)}^t \right\|^2 \mid \mathcal{X}_{(n,m)}^t \subseteq \mathcal{B}^t \right\} \right)}_{\text{IV}}.
 \end{aligned} \tag{50}$$

For I, with Assumption 4, we have

$$\text{I} = -\eta \left\langle \nabla_{\mathbf{w}_{\text{nc}}} F(\mathbf{w}^t, \mathcal{D}), \mathbb{E}_{\mathcal{B}^t} \left\{ \nabla_{\mathbf{w}_{\text{nc}}} F(\mathbf{w}^t, \mathcal{B}^t) \right\} \right\rangle = -\eta \|\nabla_{\mathbf{w}_{\text{nc}}} F(\mathbf{w}^t, \mathcal{D})\|^2. \tag{51}$$

For II, with Assumption 5, we have

$$\begin{aligned}
 \text{II} &= \frac{L_1}{2} \mathbb{E}_{\mathcal{B}^t} \left\{ \left\| -\eta \nabla_{\mathbf{w}_{\text{nc}}} F(\mathbf{w}^t, \mathcal{B}^t) \right\|^2 \right\} \\
 &= \frac{\eta^2 L_1}{2} \mathbb{E}_{\mathcal{B}^t} \left\{ \left\| \nabla_{\mathbf{w}_{\text{nc}}} F(\mathbf{w}^t, \mathcal{B}^t) - \nabla_{\mathbf{w}_{\text{nc}}} F(\mathbf{w}^t, \mathcal{D}) + \nabla_{\mathbf{w}_{\text{nc}}} F(\mathbf{w}^t, \mathcal{D}) \right\|^2 \right\} \\
 &= \frac{\eta^2 L_1}{2} \left(\underbrace{\mathbb{E}_{\mathcal{B}^t} \left\{ \left\| \nabla_{\mathbf{w}_{\text{nc}}} F(\mathbf{w}^t, \mathcal{B}^t) - \nabla_{\mathbf{w}_{\text{nc}}} F(\mathbf{w}^t, \mathcal{D}) \right\|^2 \right\}}_{\text{V}} + \|\nabla_{\mathbf{w}_{\text{nc}}} F(\mathbf{w}^t, \mathcal{D})\|^2 \right),
 \end{aligned} \tag{52}$$

where

$$\text{V} = \mathbb{E}_{\mathcal{B}^t} \left\{ \left\| \frac{1}{B} \sum_{\mathbf{x}_i \in \mathcal{B}^t} (f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t) - \nabla_{\mathbf{w}_{\text{nc}}} F(\mathbf{w}^t, \mathcal{D})) \right\|^2 \right\}$$

$$\begin{aligned}
 &= \frac{1}{B^2} \left(\sum_{\mathbf{x}_i \in \mathcal{B}^t} \mathbb{E}_{\mathcal{B}^t} \left\{ \left\| f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t) - \nabla_{\mathbf{w}_{\text{nc}}} F(\mathbf{w}^t, \mathcal{D}) \right\|^2 \right\} \right. \\
 &\quad \left. + 2 \sum_{\mathbf{x}_i, \mathbf{x}_j \in \mathcal{B}^t (i \neq j)} \mathbb{E}_{\mathcal{B}^t} \left\{ \left\langle f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t) - \nabla_{\mathbf{w}_{\text{nc}}} F(\mathbf{w}^t, \mathcal{D}), f(\mathbf{w}^t, \mathbf{x}_j, \mathcal{A}_j^t) - \nabla_{\mathbf{w}_{\text{nc}}} F(\mathbf{w}^t, \mathcal{D}) \right\rangle \right\} \right). \quad (53)
 \end{aligned}$$

With Assumption 4, we have $\mathbb{E}_{\mathbf{x}_i} \{ \nabla_{\mathbf{w}_{\text{nc}}} f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t) \} - \nabla_{\mathbf{w}_{\text{nc}}} F(\mathbf{w}^t, \mathcal{D}) = 0$, thus, we rewrite V as

$$\begin{aligned}
 \text{V} &= \frac{1}{B^2} \sum_{\mathbf{x}_i \in \mathcal{B}^t} \mathbb{E}_{\mathbf{x}_i} \left\{ \left\| \nabla_{\mathbf{w}_{\text{nc}}} f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t) - \nabla_{\mathbf{w}_{\text{nc}}} F(\mathbf{w}^t, \mathcal{D}) \right\|^2 \right\} \\
 &\leq \frac{1}{B} (\mathbb{E}_{\mathbf{x}_i} \{ \left\| \nabla_{\mathbf{w}_{\text{nc}}} f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t) \right\|^2 \} - \left\| \nabla_{\mathbf{w}_{\text{nc}}} F(\mathbf{w}^t, \mathcal{D}) \right\|^2) \\
 &\leq \frac{1}{B} (\beta \left\| \nabla_{\mathbf{w}_{\text{nc}}} F(\mathbf{w}^t, \mathcal{D}) \right\|^2 + \sigma^2 - \left\| \nabla_{\mathbf{w}_{\text{nc}}} F(\mathbf{w}^t, \mathcal{D}) \right\|^2) \\
 &= \frac{1}{B} ((\beta - 1) \left\| \nabla_{\mathbf{w}_{\text{nc}}} F(\mathbf{w}^t, \mathcal{D}) \right\|^2 + \sigma^2). \quad (54)
 \end{aligned}$$

Consequently, we have

$$\begin{aligned}
 \text{II} &\leq \frac{\eta^2 L_1}{2} \left(\frac{1}{B} ((\beta - 1) \left\| \nabla_{\mathbf{w}_{\text{nc}}} F(\mathbf{w}^t, \mathcal{D}) \right\|^2 + \sigma^2) + \left\| \nabla_{\mathbf{w}_{\text{nc}}} F(\mathbf{w}^t, \mathcal{D}) \right\|^2 \right) \\
 &= \frac{\eta^2 L_1}{2} \left(\left(1 + \frac{\beta - 1}{B} \right) \left\| \nabla_{\mathbf{w}_{\text{nc}}} F(\mathbf{w}^t, \mathcal{D}) \right\|^2 + \frac{\sigma^2}{B} \right). \quad (55)
 \end{aligned}$$

For III, with Assumption 4, we have

$$\begin{aligned}
 \text{III} &= -\eta \sum_{(n,m) \in \mathcal{E}} \frac{\tilde{X}_{(n,m)}^t}{D} \frac{1}{\tilde{X}_{(n,m)}^t} \sum_{\mathbf{x}_i \in \tilde{\mathcal{X}}_{(n,m)}^t} \left\langle \nabla_{\mathbf{w}_{(n,m)}} f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t), \mathbb{E}_{\mathcal{X}_{(n,m)}^t} \left\{ \nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \mathcal{X}_{(n,m)}^t) \mid \mathcal{X}_{(n,m)}^t \subseteq \mathcal{B}^t \right\} \right\rangle \\
 &= -\eta \sum_{(n,m) \in \mathcal{E}} \frac{\tilde{X}_{(n,m)}^t}{D} \left\langle \nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \tilde{X}_{(n,m)}^t), \nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \tilde{\mathcal{X}}_{(n,m)}^t) \right\rangle \\
 &= -\eta \sum_{(n,m) \in \mathcal{E}} \frac{\tilde{X}_{(n,m)}^t}{D} \left\| \nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \tilde{X}_{(n,m)}^t) \right\|^2. \quad (56)
 \end{aligned}$$

For IV, with Assumption 5, we have

$$\begin{aligned}
 \text{IV} &= \sum_{(n,m) \in \mathcal{E}} \frac{\tilde{X}_{(n,m)}^t}{D} \frac{1}{\tilde{X}_{(n,m)}^t} \sum_{\mathbf{x}_i \in \tilde{\mathcal{X}}_{(n,m)}^t} \frac{L_2 N K}{2} \mathbb{E}_{\mathcal{X}_{(n,m)}^t} \left\{ \left\| -\eta \nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \mathcal{X}_{(n,m)}^t) \right\|^2 \mid \mathcal{X}_{(n,m)}^t \subseteq \mathcal{B}^t \right\} \\
 &= \sum_{(n,m) \in \mathcal{E}} \frac{\tilde{X}_{(n,m)}^t}{D} \frac{\eta^2 L_2 N K}{2} \\
 &\quad \times \mathbb{E}_{\mathcal{X}_{(n,m)}^t} \left\{ \left\| \nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \mathcal{X}_{(n,m)}^t) - \nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \tilde{\mathcal{X}}_{(n,m)}^t) + \nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \tilde{\mathcal{X}}_{(n,m)}^t) \right\|^2 \mid \mathcal{X}_{(n,m)}^t \subseteq \mathcal{B}^t \right\} \\
 &= \sum_{(n,m) \in \mathcal{E}} \frac{\tilde{X}_{(n,m)}^t}{D} \frac{\eta^2 L_2 N K}{2} \\
 &\quad \times \underbrace{\left(\mathbb{E}_{\mathcal{X}_{(n,m)}^t} \left\{ \left\| \nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \mathcal{X}_{(n,m)}^t) - \nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \tilde{\mathcal{X}}_{(n,m)}^t) \right\|^2 \mid \mathcal{X}_{(n,m)}^t \subseteq \mathcal{B}^t \right\} \right)}_{\text{VI}} + \left\| \nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \tilde{\mathcal{X}}_{(n,m)}^t) \right\|^2 \quad (57)
 \end{aligned}$$

where

$$\begin{aligned}
 \text{VI} &= \mathbb{E}_{\mathcal{X}_{(n,m)}^t} \left\{ \left\| \frac{1}{X_{(n,m)}^t} \sum_{\mathbf{x}_i \in \mathcal{X}_{(n,m)}^t} \left(\nabla_{\mathbf{w}_{(n,m)}} f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t) - \nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \tilde{\mathcal{X}}_{(n,m)}^t) \right) \right\|^2 \middle| \mathcal{X}_{(n,m)}^t \subseteq \mathcal{B}^t \right\} \\
 &= \frac{1}{(X_{(n,m)}^t)^2} \left(\sum_{\mathbf{x}_i \in \mathcal{X}_{(n,m)}^t} \mathbb{E}_{\mathcal{X}_{(n,m)}^t} \left\{ \left\| \nabla_{\mathbf{w}_{(n,m)}} f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t) - \nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \tilde{\mathcal{X}}_{(n,m)}^t) \right\|^2 \middle| \mathcal{X}_{(n,m)}^t \subseteq \mathcal{B}^t \right\} \right. \\
 &\quad \left. + 2 \sum_{\mathbf{x}_i, \mathbf{x}_j \in \mathcal{X}_{(n,m)}^t (i \neq j)} \mathbb{E}_{\mathcal{X}_{(n,m)}^t} \left\{ \left\langle \nabla_{\mathbf{w}_{(n,m)}} f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t) - \nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \tilde{\mathcal{X}}_{(n,m)}^t), \right. \right. \\
 &\quad \left. \left. \nabla_{\mathbf{w}_{(n,m)}} f(\mathbf{w}^t, \mathbf{x}_j, \mathcal{A}_j^t) - \nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \tilde{\mathcal{X}}_{(n,m)}^t) \right\rangle \middle| \mathcal{X}_{(n,m)}^t \subseteq \mathcal{B}^t \right\} \right). \tag{58}
 \end{aligned}$$

With Assumption 4, we have $\mathbb{E}_{\mathcal{X}_{(n,m)}^t} \left\{ \nabla_{\mathbf{w}_{(n,m)}} f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t) \middle| \mathcal{X}_{(n,m)}^t \subseteq \mathcal{B}^t \right\} - \nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \tilde{\mathcal{X}}_{(n,m)}^t) = 0$, thus, we rewrite VI as

$$\begin{aligned}
 \text{VI} &= \frac{1}{(X_{(n,m)}^t)^2} \sum_{\mathbf{x}_i \in \mathcal{X}_{(n,m)}^t} \mathbb{E}_{\mathcal{X}_{(n,m)}^t} \left\{ \left\| \nabla_{\mathbf{w}_{(n,m)}} f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t) - \nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \tilde{\mathcal{X}}_{(n,m)}^t) \right\|^2 \middle| \mathcal{X}_{(n,m)}^t \subseteq \mathcal{B}^t \right\} \\
 &= \frac{1}{X_{(n,m)}^t} \left(\mathbb{E}_{\mathbf{x}_i} \left\{ \left\| \nabla_{\mathbf{w}_{(n,m)}} f(\mathbf{w}^t, \mathbf{x}_i, \mathcal{A}_i^t) \right\|^2 \middle| \mathbf{x}_i \in \mathcal{X}_{(n,m)}^t \subseteq \mathcal{B}^t \right\} - \left\| \nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \tilde{\mathcal{X}}_{(n,m)}^t) \right\|^2 \right) \\
 &= \frac{1}{X_{(n,m)}^t} \left((\beta_{(n,m)} - 1) \left\| \nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \tilde{\mathcal{X}}_{(n,m)}^t) \right\|^2 + \sigma_{(n,m)}^2 \right). \tag{59}
 \end{aligned}$$

Consequently, we have

$$\begin{aligned}
 \text{IV} &\leq \sum_{(n,m) \in \mathcal{E}} \frac{\tilde{X}_{(n,m)}^t}{D} \frac{\eta^2 L_2 N K}{2} \left(\frac{\beta_{(n,m)} - 1}{X_{(n,m)}^t} \left\| \nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \tilde{\mathcal{X}}_{(n,m)}^t) \right\|^2 + \frac{\sigma_{(n,m)}^2}{X_{(n,m)}^t} + \left\| \nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \tilde{\mathcal{X}}_{(n,m)}^t) \right\|^2 \right) \\
 &= \sum_{(n,m) \in \mathcal{E}} \frac{\tilde{X}_{(n,m)}^t}{D} \frac{\eta^2 L_2 N K}{2} \left(\left(1 + \frac{\beta_{(n,m)} - 1}{X_{(n,m)}^t} \right) \left\| \nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \tilde{\mathcal{X}}_{(n,m)}^t) \right\|^2 + \frac{\sigma_{(n,m)}^2}{X_{(n,m)}^t} \right). \tag{60}
 \end{aligned}$$

Combining the results of I, II, III, and IV, we have

$$\begin{aligned}
 &\mathbb{E}_{\mathcal{B}^t} \left\{ \tilde{F}(\mathbf{w}^{t+1}, \mathcal{D}) \right\} - F(\mathbf{w}^t, \mathcal{D}) \\
 &\leq - \underbrace{\eta \left(1 - \frac{\eta L_1 (B - 1 + \beta)}{2B} \right)}_{\phi} \left\| \nabla_{\mathbf{w}_{\text{nc}}} F(\mathbf{w}^t, \mathcal{D}) \right\|^2 + \frac{\eta^2 L_1}{2} \frac{\sigma^2}{B} \\
 &\leq - \sum_{(n,m) \in \mathcal{E}} \underbrace{\frac{\eta \tilde{X}_{(n,m)}^t}{D} \left(1 - \frac{\eta L_2 N K (X_{(n,m)}^t - 1 + \beta_{(n,m)})}{2 X_{(n,m)}^t} \right)}_{\phi_{(n,m)}^t} \left\| \nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \tilde{\mathcal{X}}_{(n,m)}^t) \right\|^2 \\
 &\quad + \sum_{(n,m) \in \mathcal{E}} \frac{\tilde{X}_{(n,m)}^t}{D} \frac{\eta^2 L_2 N K \sigma_{(n,m)}^2}{X_{(n,m)}^t}. \tag{61}
 \end{aligned}$$

If

$$\eta < \eta_0 < \min \left\{ \frac{B}{L_1 (B - 1 + \beta)}, \frac{\left\| \nabla_{\mathbf{w}_{\text{nc}}} F(\mathbf{w}^t, \mathcal{D}) \right\|^2 B}{L_1 \sigma^2} \right\},$$

$$\min_{(n,m) \in \mathcal{E}, t \in \mathcal{T}} \left\{ \frac{X_{(n,m)}^t}{L_2 N K (X_{(n,m)}^t - 1 + \beta_{(n,m)})} \right\}, \min_{(n,m) \in \mathcal{E}, t \in \mathcal{T}} \left\{ \frac{\|\nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \tilde{\mathcal{X}}_{(n,m)}^t)\|^2 X_{(n,m)}^t}{L_2 N K \sigma_{(n,m)}^2} \right\}, \quad (62)$$

where η_0 is a constant, we have $\phi > \frac{1}{2}$ and $\phi_{(n,m)}^t > \frac{1}{2}$ ($\forall (n,m) \in \mathcal{E}, t \in \mathcal{T}$). Moreover, if $\mathbb{E} \left\{ \frac{\tilde{X}_{(n,m)}^t}{D} \right\} \in [\tilde{\mu}_1, \tilde{\mu}_2]$ and $\mathbb{E} \left\{ \frac{X_{(n,m)}^t}{B} \right\} \in [\mu_1, \mu_2]$, where $0 < \tilde{\mu}_1 \leq \tilde{\mu}_2 < 1$ and $0 < \mu_1 \leq \mu_2 < 1$, we rearrange the terms in the above inequalities to obtain

$$\mathbb{E} \{F(\mathbf{w}^{t+1}, \mathcal{D})\} - \mathbb{E} \{F(\mathbf{w}^t, \mathcal{D})\} \leq -\frac{\eta}{2} \mathbb{E} \{\|\nabla_{\mathbf{w}_{\text{nc}}} F(\mathbf{w}^t, \mathcal{D})\|^2\} + \frac{\eta^2 L_1 \sigma^2}{2B} + 4N\Xi \left(1 - \frac{K}{M}\right) \mathbb{E} \{\delta^t\}, \quad (63)$$

and

$$\begin{aligned} \mathbb{E} \{F(\mathbf{w}^{t+1}, \mathcal{D})\} - \mathbb{E} \{F(\mathbf{w}^t, \mathcal{D})\} &\leq -\frac{\eta}{2} \sum_{(n,m) \in \mathcal{E}} \mathbb{E} \left\{ \frac{\tilde{X}_{(n,m)}^t}{D} \|\nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \tilde{\mathcal{X}}_{(n,m)}^t)\|^2 \right\} \\ &+ \sum_{(n,m) \in \mathcal{E}} \mathbb{E} \left\{ \frac{\tilde{X}_{(n,m)}^t}{D} \frac{\eta^2 L_2 N K \sigma_{(n,m)}^2}{X_{(n,m)}^t} \right\} + 4N\Xi \left(1 - \frac{K}{M}\right) \mathbb{E} \{\delta^t\} \\ &\leq -\frac{\tilde{\mu}_1 \eta}{2} \sum_{(n,m) \in \mathcal{E}} \mathbb{E} \{\|\nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \tilde{\mathcal{X}}_{(n,m)}^t)\|^2\} + \frac{\sum_{(n,m) \in \mathcal{E}} \tilde{\mu}_2 \eta^2 L_2 N K \sigma_{(n,m)}^2}{2\mu_1 B} + 4N\Xi \left(1 - \frac{K}{M}\right) \mathbb{E} \{\delta^t\}. \end{aligned} \quad (64)$$

For (63), we sum over $t = 0$ to $T - 1$ and rearrange the terms to obtain

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \{\|\nabla_{\mathbf{w}_{\text{nc}}} F(\mathbf{w}^t, \mathcal{D})\|^2\} \leq \frac{2(F(\mathbf{w}^0, \mathcal{D}) - F_{\text{inf}})}{\eta T} + \frac{\eta L_1 \sigma^2}{B} + \frac{8N\Xi}{\eta T} \left(1 - \frac{K}{M}\right) \sum_{t=0}^{T-1} \mathbb{E} \{\delta^t\}. \quad (65)$$

For (64), we sum over $t = 0$ to $T - 1$ and rearrange the terms to obtain

$$\begin{aligned} \frac{1}{T} \sum_{t=0}^{T-1} \sum_{(n,m) \in \mathcal{E}} \mathbb{E} \left\{ \|\nabla_{\mathbf{w}_{(n,m)}} F(\mathbf{w}^t, \tilde{\mathcal{X}}_{(n,m)}^t)\|^2 \right\} &\leq \frac{2(F(\mathbf{w}^0, \mathcal{D}) - F_{\text{inf}})}{\tilde{\mu}_1 \eta T} \\ &+ \frac{\sum_{(n,m) \in \mathcal{E}} \tilde{\mu}_2 \eta L_2 N K \sigma_{(n,m)}^2}{\tilde{\mu}_1 \mu_1 B} + \frac{8N\Xi}{\tilde{\mu}_1} \left(1 - \frac{K}{M}\right) \frac{1}{\eta T} \sum_{t=0}^{T-1} \mathbb{E} \{\delta^t\}. \end{aligned} \quad (66)$$

Finally, we sum (65) and (66):

$$\begin{aligned} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \{\|\nabla_{\mathbf{w}} F(\mathbf{w}^t, \mathcal{D})\|^2\} &\leq \frac{2(F(\mathbf{w}^0, \mathcal{D}) - F_{\text{inf}})}{\eta T} \left(1 + \frac{1}{\tilde{\mu}_1}\right) + \frac{\eta L_1 \sigma^2}{B} \\ &+ \frac{\sum_{(n,m) \in \mathcal{E}} \tilde{\mu}_2 \eta L_2 N K \sigma_{(n,m)}^2}{\tilde{\mu}_1 \mu_1 B} + \frac{8N\Xi}{\eta} \left(1 - \frac{K}{M}\right) \left(1 + \frac{1}{\tilde{\mu}_1}\right) \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \{\delta^t\}. \end{aligned} \quad (67)$$

Now, let us bound the term $\sum_{t=0}^{T-1} \mathbb{E} \{\delta^t\}$. If $\mathbb{E} \{F(\mathbf{w}^{t+1}, \mathcal{D})\} - \mathbb{E} \{F(\mathbf{w}^t, \mathcal{D})\} \leq \frac{4\Xi N(M-K)\delta^t}{M} \leq \mathbb{E} \{F(\mathbf{w}^t, \mathcal{D})\} - \mathbb{E} \{F(\mathbf{w}^{t+1}, \mathcal{D})\}$, we have

$$\begin{aligned} \sum_{t=0}^{T-1} \frac{4\Xi N(M-K)\mathbb{E} \{\delta^t\}}{M} &\leq \mathbb{E} \left\{ \sum_{t=0}^{T-1} (F(\mathbf{w}^t, \mathcal{D}) - F(\mathbf{w}^{t+1}, \mathcal{D})) \right\} \\ &= F(\mathbf{w}^0, \mathcal{D}) - F(\mathbf{w}^{T-1}, \mathcal{D}) \leq F(\mathbf{w}^0, \mathcal{D}) - F_{\text{inf}}, \end{aligned} \quad (68)$$

which indicates that $\sum_{t=0}^{T-1} \mathbb{E} \{\delta^t\}$ is upper bounded by a constant (denoted by $\tilde{\delta}$):

$$\sum_{t=0}^{T-1} \mathbb{E} \{\delta^t\} \leq \tilde{\delta}. \quad (69)$$

Combining the above results, we complete the proof. $\square$

D. Proof of Corollary 2

Corollary 2. *If $\left\{ \frac{\tilde{X}_{(n,m)}^t}{D}, \frac{X_{(n,m)}^t}{B} \right\} = \Theta\left(\frac{K}{M}\right)$, $\forall (n, m) \in \mathcal{E}, \forall t \in \mathcal{T}$ and let $\eta = \Theta\left(\frac{\eta_0}{\sqrt{T}}\right)$ hold, the asymptotic convergence bound of the the average gradient norm of MoE over all iterations is given by*

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \{\|\nabla_{\mathbf{w}} F(\mathbf{w}^t, \mathcal{D})\|^2\} \leq \mathcal{O} \left(\left(Z_1 + Z_3 \left(1 - \frac{K}{M} \right) \right) \frac{1 + \frac{M}{K}}{\sqrt{T}} \right) + \mathcal{O} \left(\frac{Z_2}{\sqrt{T}} \right), \quad (70)$$

where $Z_1 = \frac{F(\mathbf{w}^0, \mathcal{D}) - F_{\inf}}{\eta_0}$, $Z_2 = \frac{\eta_0}{B} \left(NM \sum_{(n,m) \in \mathcal{E}} \sigma_{(n,m)}^2 + \sigma^2 \right)$, and $Z_3 = \frac{8N\Xi\tilde{\delta}}{\eta_0}$.

Proof. With $\left\{ \frac{\tilde{X}_{(n,m)}^t}{D}, \frac{X_{(n,m)}^t}{B} \right\} = \Theta\left(\frac{K}{M}\right)$ ($\forall (n, m) \in \mathcal{E}, t \in \mathcal{T}$) and $\eta = \Theta\left(\frac{\eta_0}{\sqrt{T}}\right)$, we have $\{\tilde{\mu}_1, \tilde{\mu}_2, \mu_1, \mu_2\} = \mathcal{O}\left(\frac{K}{M}\right)$. Substituting them into Theorem 1, we complete the proof. $\square$

E. Proof of Corollary 3

Corollary 3. *Let us define r_g , r_e and r_s (in FLOPs) as the per-token computational cost incurred by each gate, expert, and the non-MoE components, respectively. Since each token undergoes processing by a total of N gates, KN experts, and the non-MoE components, the per-token computational cost (in FLOPs) is given by*

$$r_0 = r_s + Nr_g + KNr_e. \quad (71)$$

Given the total computational budget R (in FLOPs) of MoE training, the average gradient norm of MoE over all iterations is upper bounded by

$$\begin{aligned} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \{\|\nabla_{\mathbf{w}} F(\mathbf{w}^t, \mathcal{D})\|^2\} &\leq \mathcal{O} \left(\frac{Z_2 \sqrt{Br_0}}{\sqrt{R}} \right) \\ &+ \mathcal{O} \left(\left(Z_1 + Z_3 \left(1 - \frac{K}{M} \right) \right) \frac{(1 + \frac{M}{K}) \sqrt{Br_0}}{\sqrt{R}} \right). \end{aligned} \quad (72)$$

Proof. From (25), we have $T = \frac{R}{Br_0}$. Substituting it into Corollary 2, we complete the proof. $\square$

F. Proof of Corollary 4

Corollary 4. *From Corollary 3, given total computational budget R (in FLOPs), the optimal K^* that minimizes the average gradient norm of MoE over all iterations satisfies*

$$1 \leq K^* \leq \min \left\{ \mathcal{O} \left(\frac{(Z_1 + Z_3) M}{Z_1 + Z_2 + Z_3} \right), M \right\}. \quad (73)$$

Proof. With $r_0 \leq \mathcal{O}(K)$, we rewrite (73) as

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \{\|\nabla_{\mathbf{w}} F(\mathbf{w}^t, \mathcal{D})\|^2\} \leq \mathcal{O} \left(\frac{(Z_1 + Z_3) \frac{M}{\sqrt{K}} + (Z_1 + Z_2 + Z_3) \sqrt{K}}{\sqrt{R}} \right)$$

$$= \mathcal{O} \left(\frac{(Z_1 + Z_3) M + (Z_1 + Z_2 + Z_3) K}{\sqrt{RK}} \right). \quad (74)$$

Let $\Phi = \frac{(Z_1 + Z_3)M + (Z_1 + Z_2 + Z_3)K}{\sqrt{RK}}$, the first-order derivative of Φ is given by

$$\frac{\partial \Phi}{\partial K} = \frac{(Z_1 + Z_2 + Z_3) K - (Z_1 + Z_3) M}{2\sqrt{R}(K)^{\frac{3}{2}}}. \quad (75)$$

Let $\frac{\partial \Phi}{\partial K} = 0$, we have

$$K = \frac{(Z_1 + Z_3) M}{Z_1 + Z_2 + Z_3}. \quad (76)$$

Thus, the optimal K^* that minimizes the average gradient norm of MoE over all iterations satisfies

$$1 \leq K^* \leq \min \left\{ \mathcal{O} \left(\frac{(Z_1 + Z_3) M}{Z_1 + Z_2 + Z_3} \right), M \right\}, \quad (77)$$

which completes the proof. □

G. Supplementary Material of Experiments

Used benchmark datasets:

- **CIFAR-10** is a widely used image classification dataset that consists of 60,000 32x32 color images in 10 different classes, with 6,000 images per class. There are 50,000 training images and 10,000 test images.
- **CIFAR-100** is similar to CIFAR-10 but contains 100 classes, with 600 images per class. There are 500 training images and 100 test images per class.
- **WikiText-103** is a large-scale language modeling dataset that contains over 100 million tokens extracted from verified Good and Featured articles on Wikipedia. It is commonly used for training and evaluating language models.
- **Alpaca** is a dataset for instruction-following language models, derived from self-instruct methods. It contains 52,000 instruction-response pairs generated using OpenAI’s text-davinci-003 model.
- **Dolly** is an open-source instruction-following dataset created by Databricks. It contains 15,000 high-quality instruction-response pairs generated using a combination of human annotations and model outputs.
- **GSM8K** is a benchmark dataset for evaluating the mathematical problem-solving abilities of language models. It contains 8,500 high-quality grade school math word problems that require multi-step reasoning to solve.

Used models: We design different MoE-based models for various datasets as follows.

- For CIFAR-10, we adopt a single-layer MoE-based model, consisting of a gate composed of a linear network, 8 experts each built with a two-layer CNN, a feature extractor (independent of the MoE layer) implemented via a two-layer CNN, and a linear network. The total number of parameters is approximately 3M.
- For CIFAR-100, we also use a single-layer MoE-based model. Its components include a gate based on a linear network, 16 experts each constructed with a three-layer CNN, a feature extractor (independent of the MoE layer) consisting of a three-layer CNN, and a linear network. The total number of parameters is approximately 12M.
- For WikiText-103, we employ a Transformer-based MoE-based model with 8 MoE layers. Each MoE layer comprises a gate made of a linear network and 16 experts each built with a two-layer feed-forward network, where the token dimensions is 512 and the hidden dimensions is 2048. The MoE layers are connected via self-attention mechanisms, with a total parameter count of approximately 303M.

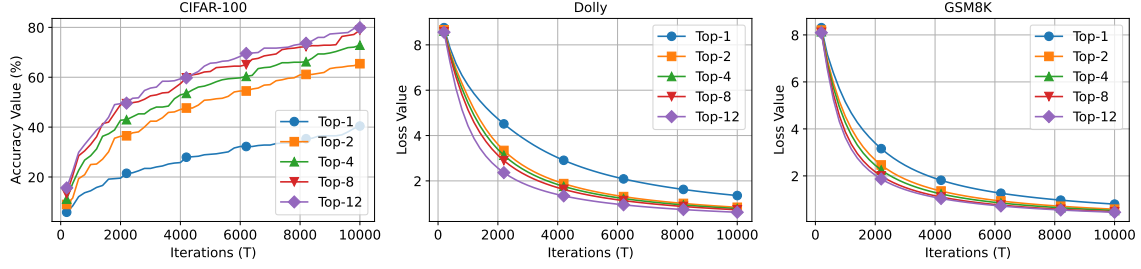


Figure 4. Loss (or Accuracy) values vs T on CIFAR-100, Dolly, and GSM8K under different K .

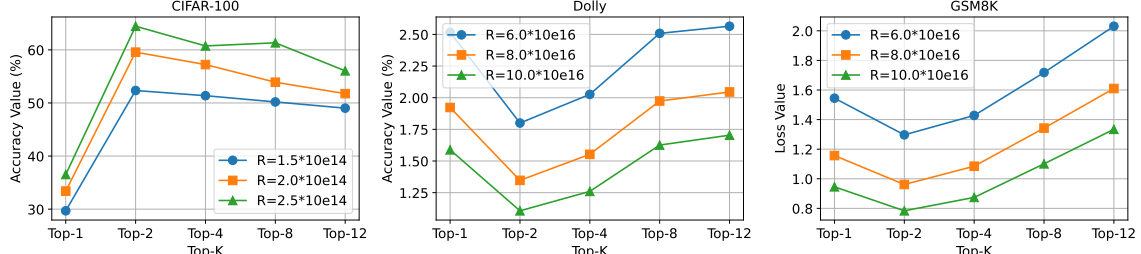


Figure 5. Loss (or Accuracy) values vs K on CIFAR-100, Dolly, and GSM8K under different R .

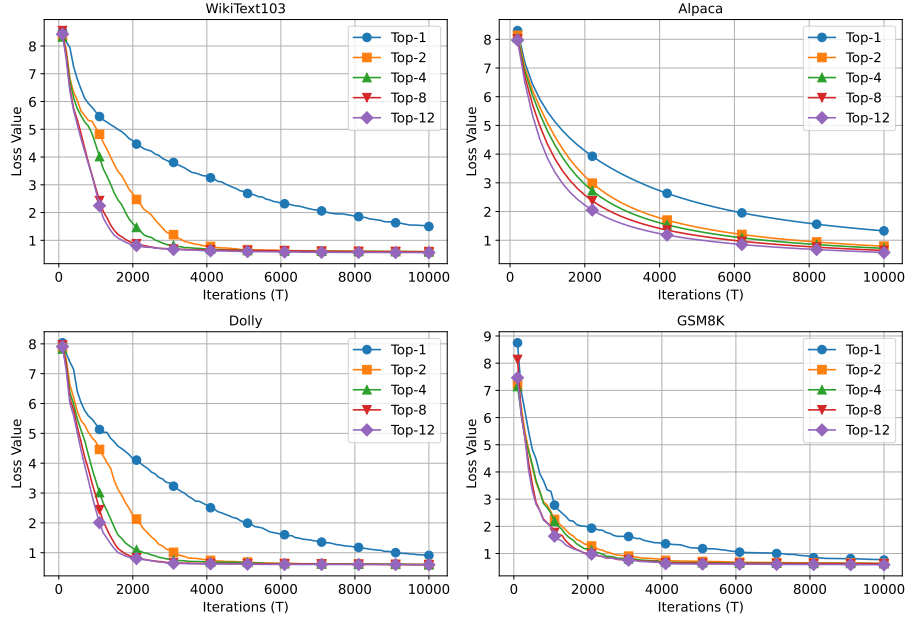


Figure 6. Loss values vs T on WikiText-103, Alpaca, Dolly, and GSM8K for large-scale MoE under different K .

- For Alpaca, Dolly, and GSM8K, we choose the model similar to that of WikiText-103, while the hidden dimensions is 1024. The total number of parameters is approximately 168M.

To further validate the theoretical analysis on larger-scale MoE models, Figs. 6 and 7 illustrate the convergence behavior under sparse routing and the convergence–cost trade-off on WikiText-103, Alpaca, Dolly, and GSM8K, respectively. Across all datasets, we employ a Transformer-based MoE model with 20 MoE layers and 16 experts per layer. The token and hidden dimensions are set to 1024 and 4096, respectively, yielding a model with approximately 3B parameters. The experimental results further demonstrate that our theoretical findings generalize well to large-scale MoE models.

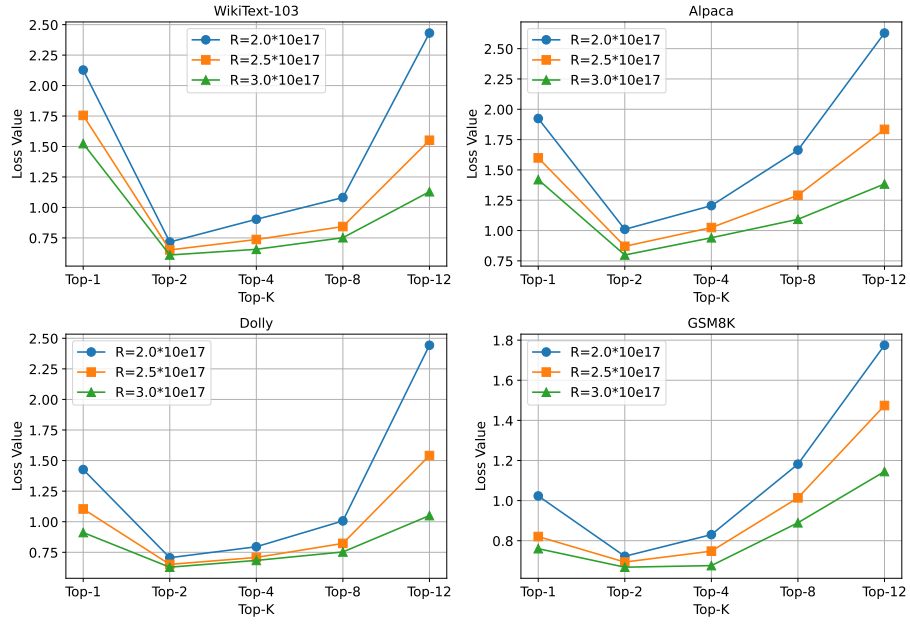


Figure 7. Loss values vs K on WikiText-103, Alpaca, Dolly, and GSM8K for large-scale MoE under different R .